Color Image Processing

6.1 Color Fundamentals

- When color is available, it gives much more information about an image than intensity alone.
- Human can discern thousands of color, compared to about only two dozen of gray.
- Color is very useful for recognition of objects in an image both for humans and computers. Color is a powerful descriptor.

Color Fundamentals

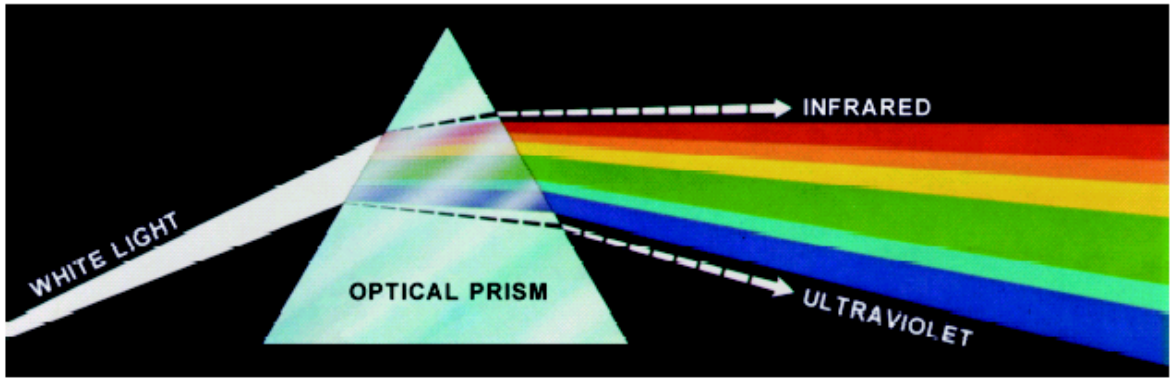


FIGURE 6.1 Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lamp Business Division.)

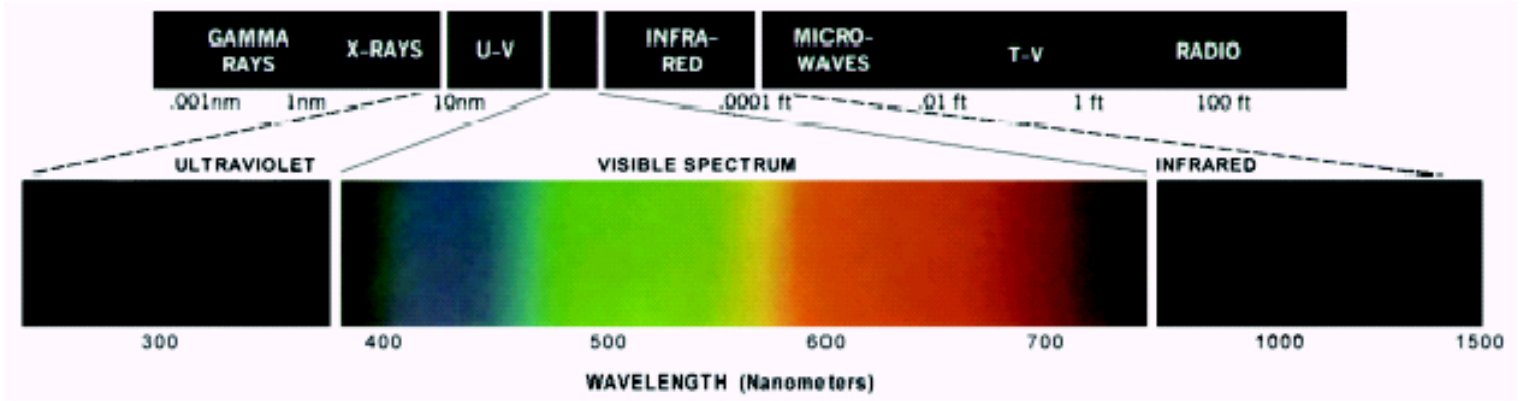
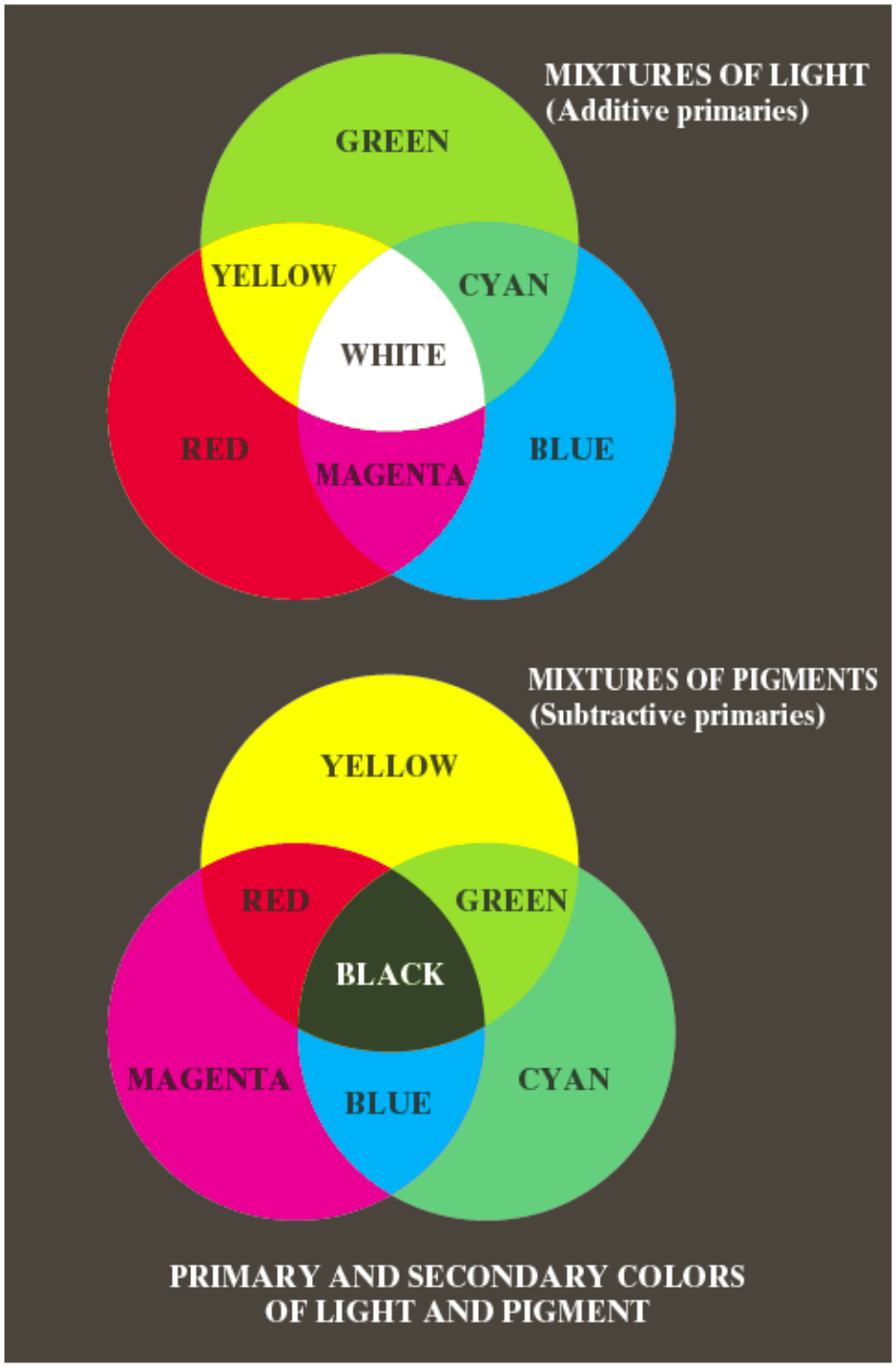


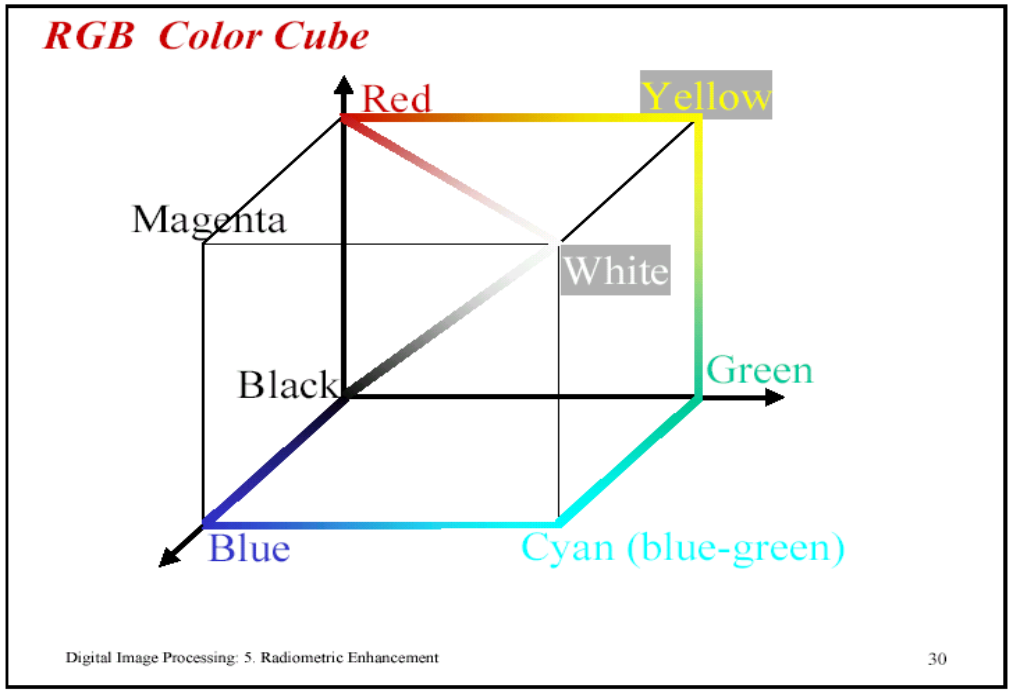
FIGURE 6.2 Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lamp Business Division.)

Color Fundamentals

- Secondary colors: magenta (red + blue), cyan (green + blue), and yellow (red + green)



6.2 Color Models -- RGB Model



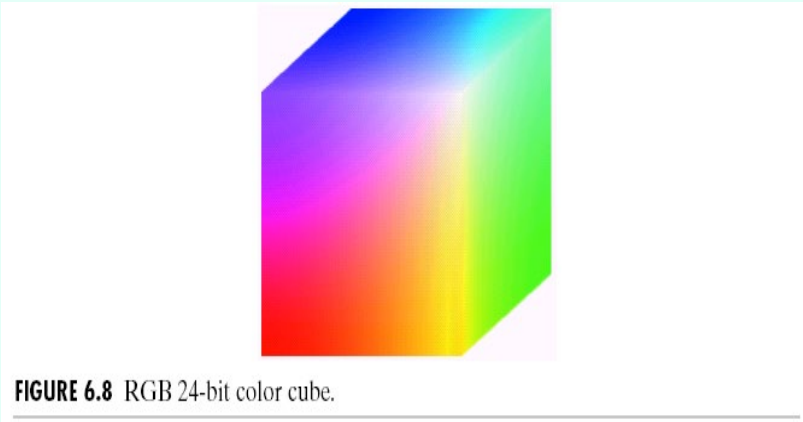


FIGURE 6.8 RGB 24-bit color cube.

Color Models -- RGB Model

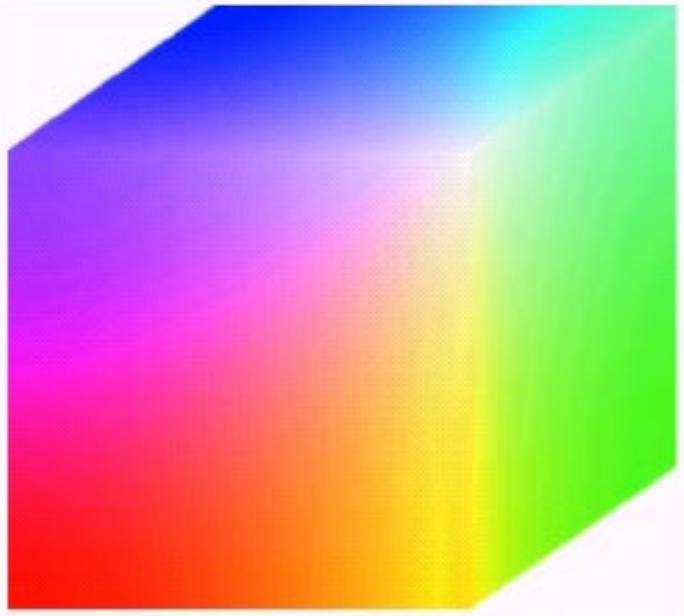
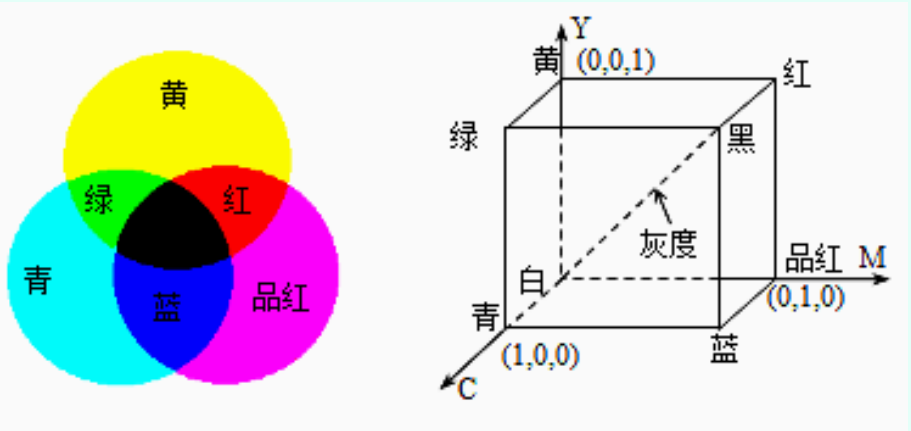
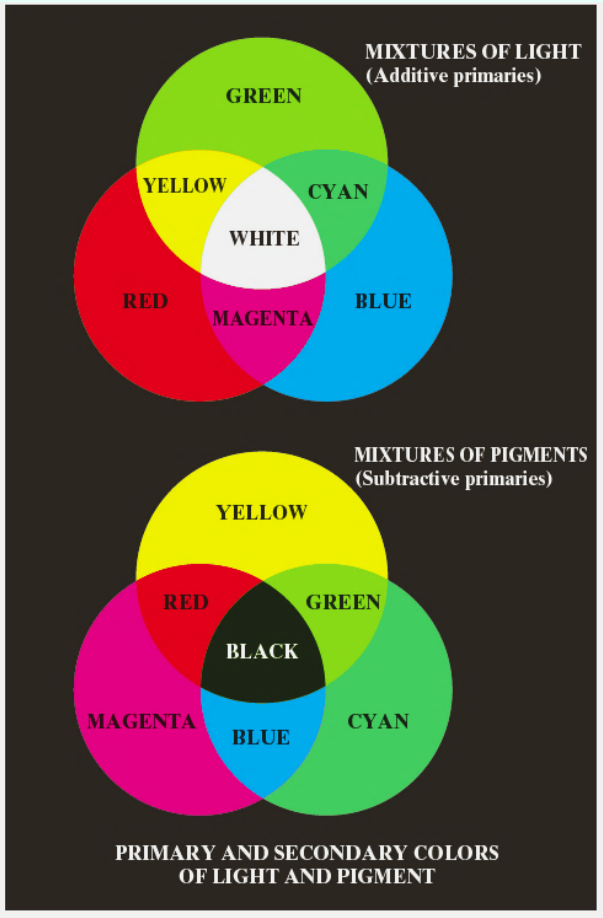


FIGURE 6.8 RGB 24-bit color cube.

Color Models -- CMY Model

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



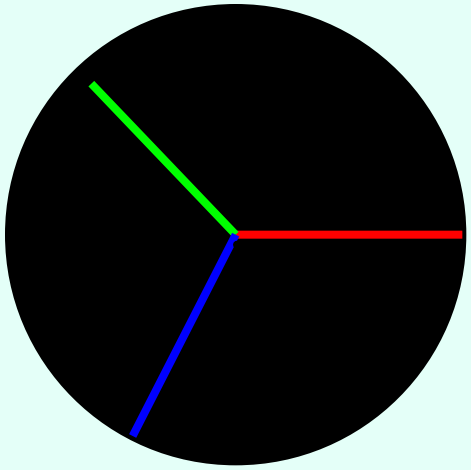


Color Models -- HSI Model

色调 hue， 饱和度 saturation， 亮度 intensity，

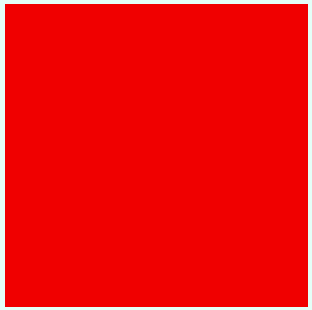


- H：denote Hue，by represented angle. 反映了该颜色最接近什么样的光谱波长。0°为红色，120°为绿色，240°为蓝色。





Red

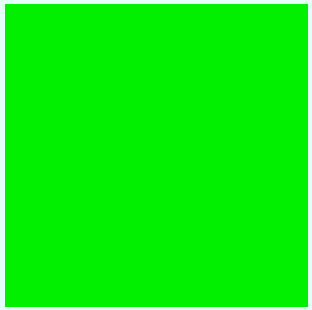


H=0º

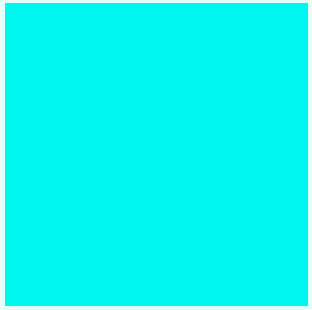
Yellow

H=60º

Green

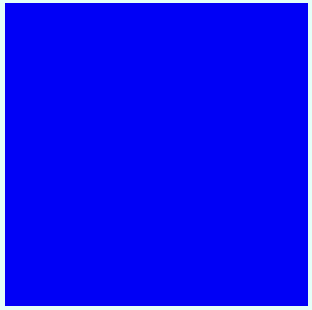


H=120º



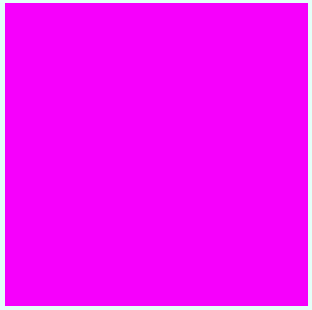
H=180º

Cyan



H=240º

Blue

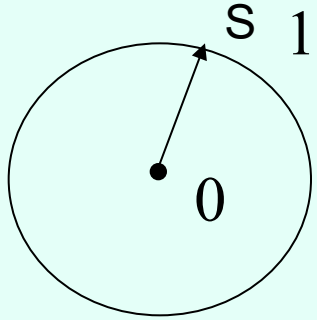


H=300º

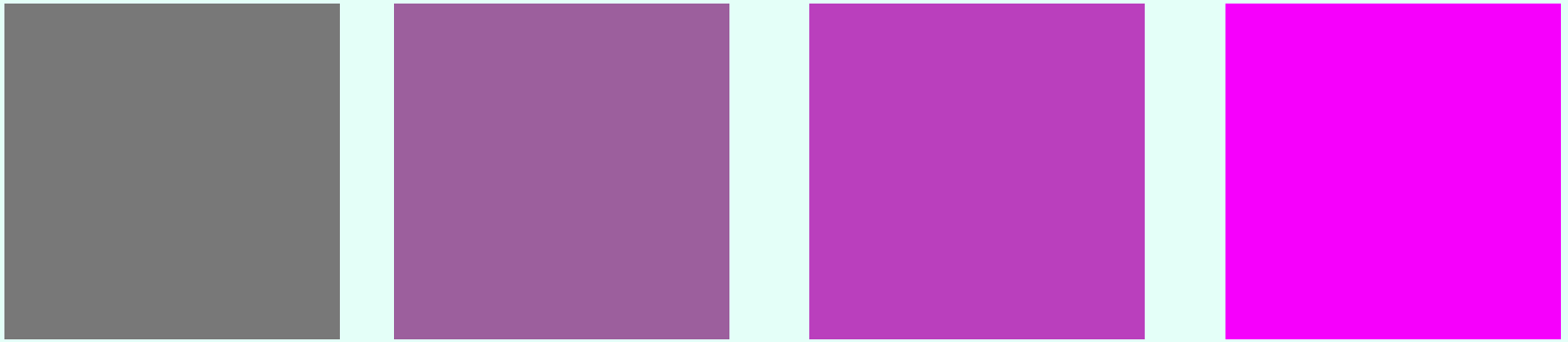
Magenta

- S：表示饱和度，饱和度参数是色环的原点到彩色点的半径长度。饱和度是指颜色的纯度，即掺入白光的程度 。

- 在环的外围圆周是纯的或称饱和的颜色，其饱和度值为1。在中心是中性（灰）色，即饱和度为0。



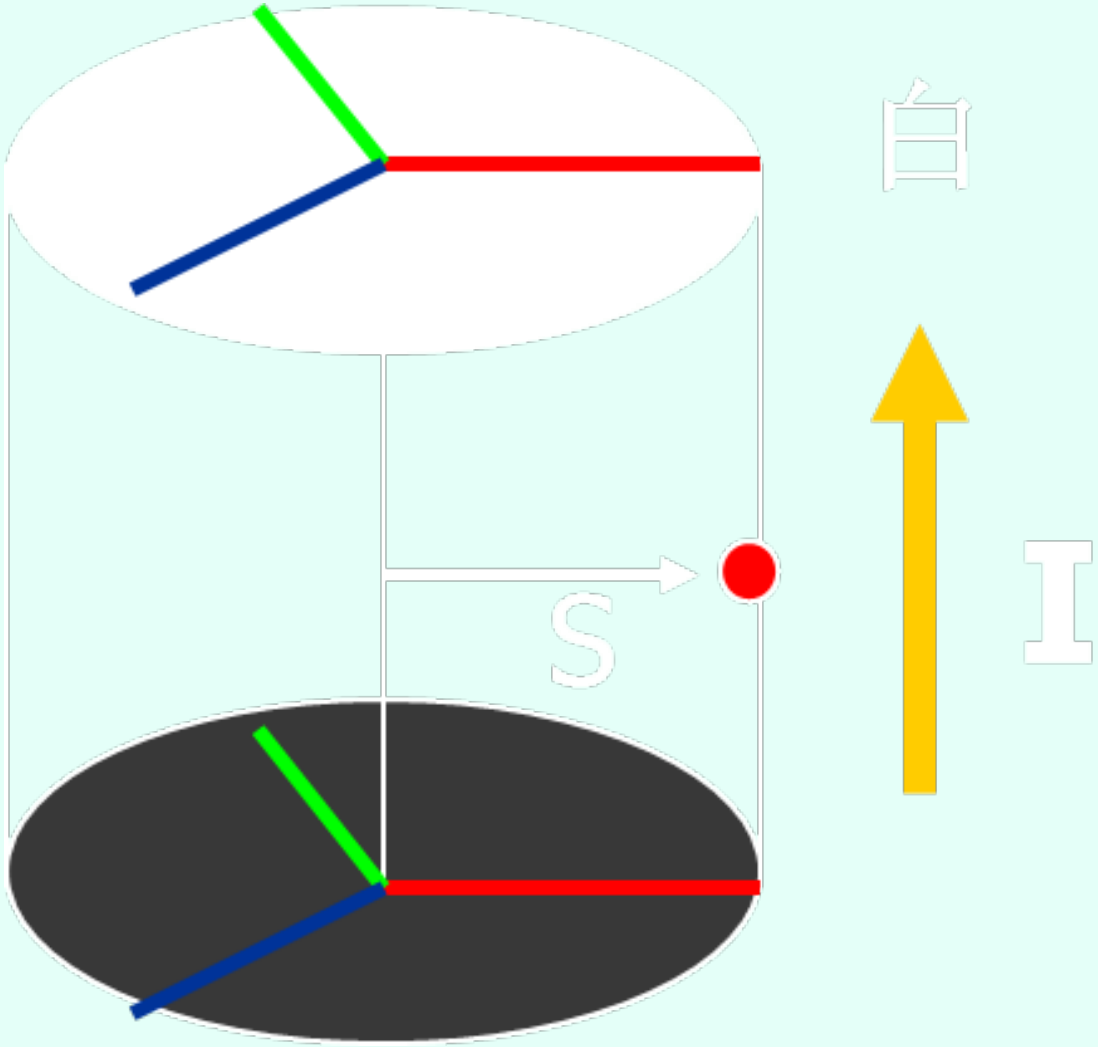
饱和度（S）的效果示意图

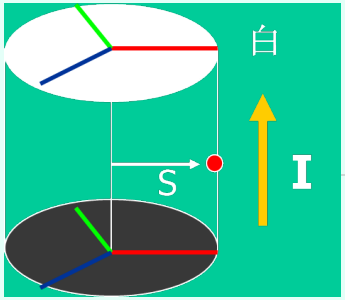


S=0　　S=1/4　　S=1/2　　S=1

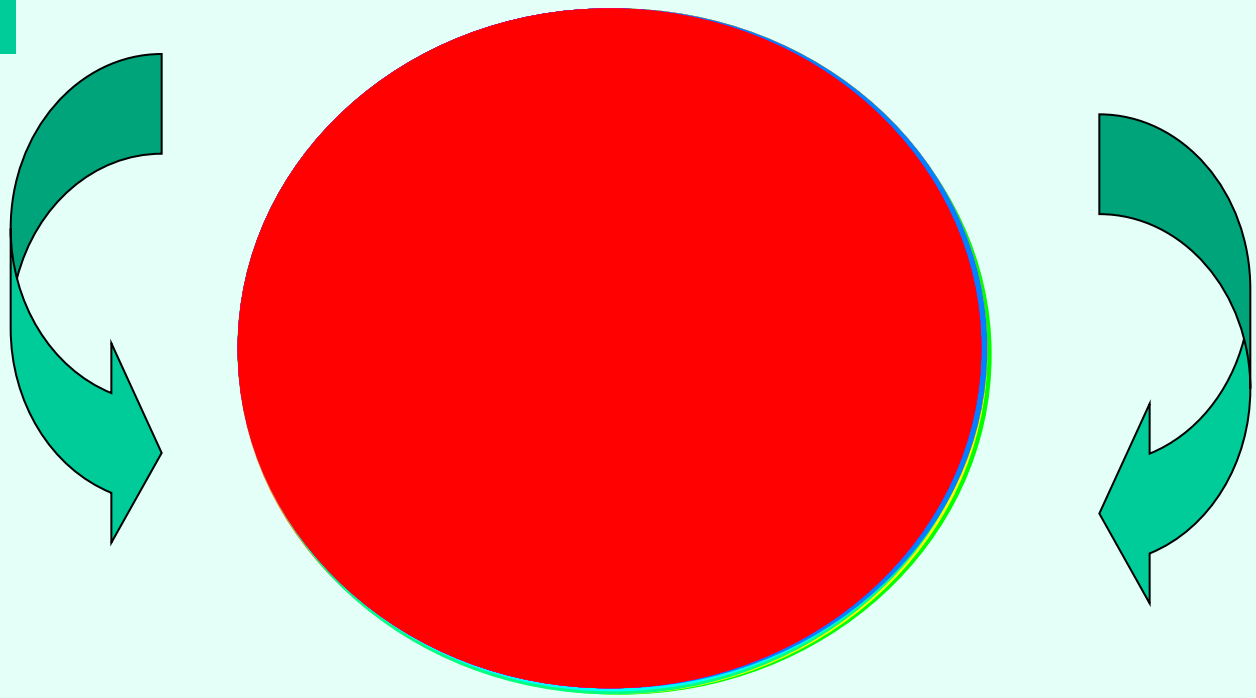
- I：**表示亮度，反映**光作用于人眼时所引起的明亮程度的感觉

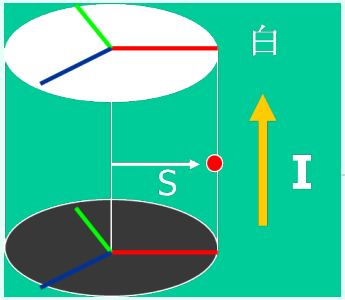
- 彩色光的强度降至使人看不清，在亮度等级上与黑色对应；同样如果其强度变得很大，那么亮度等级与白色对应。


白
S
I

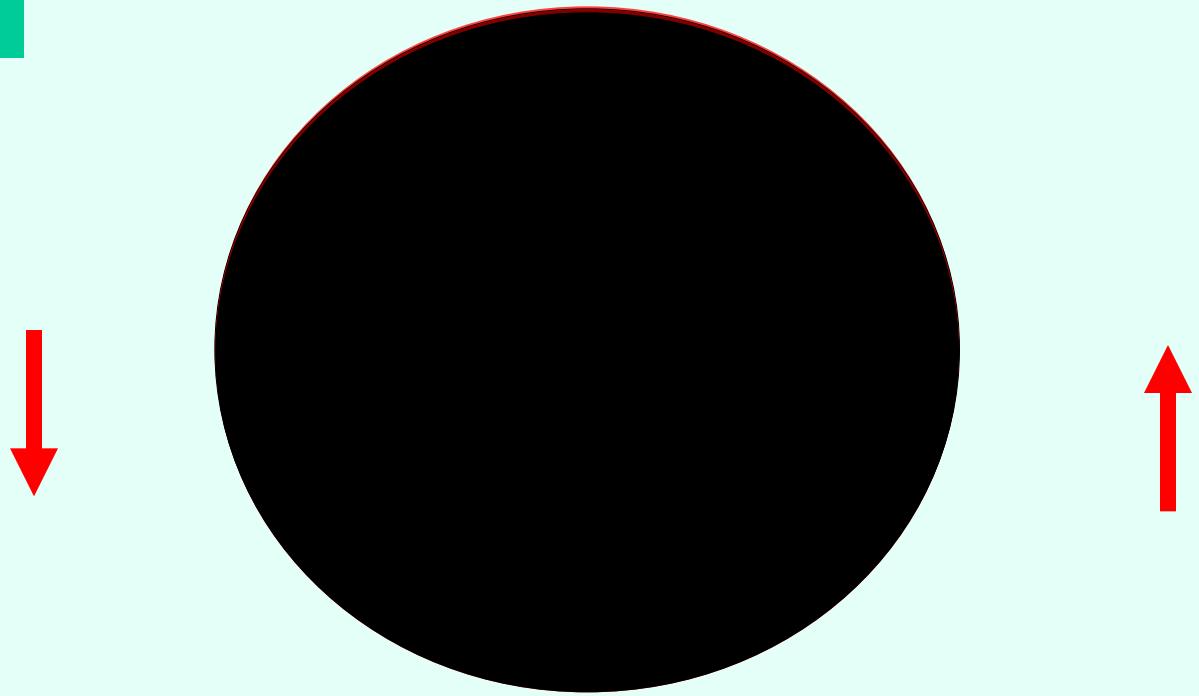


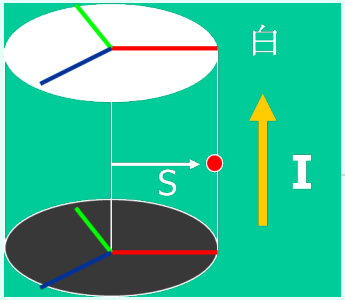
红点的顺（逆）时针转动



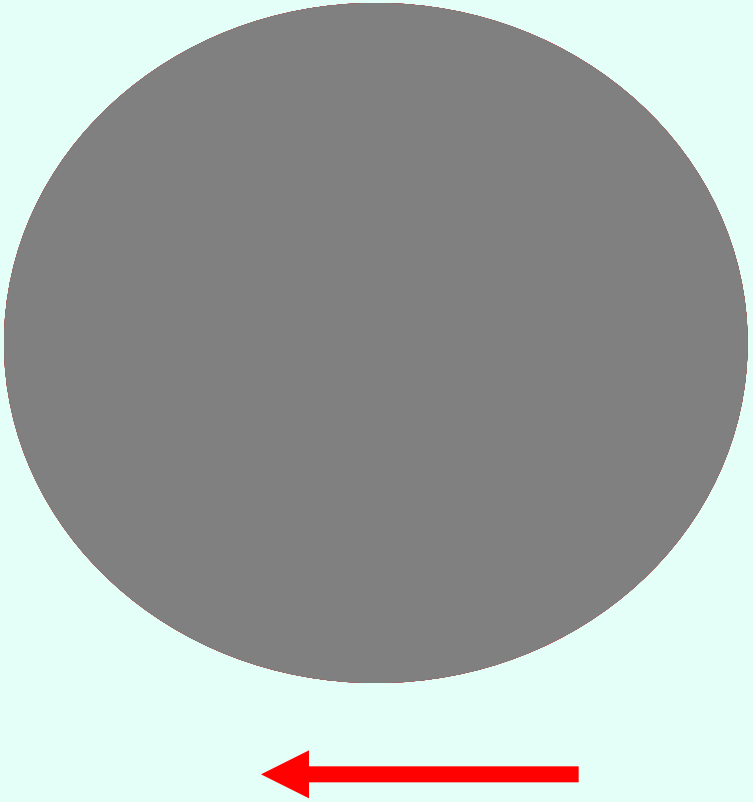


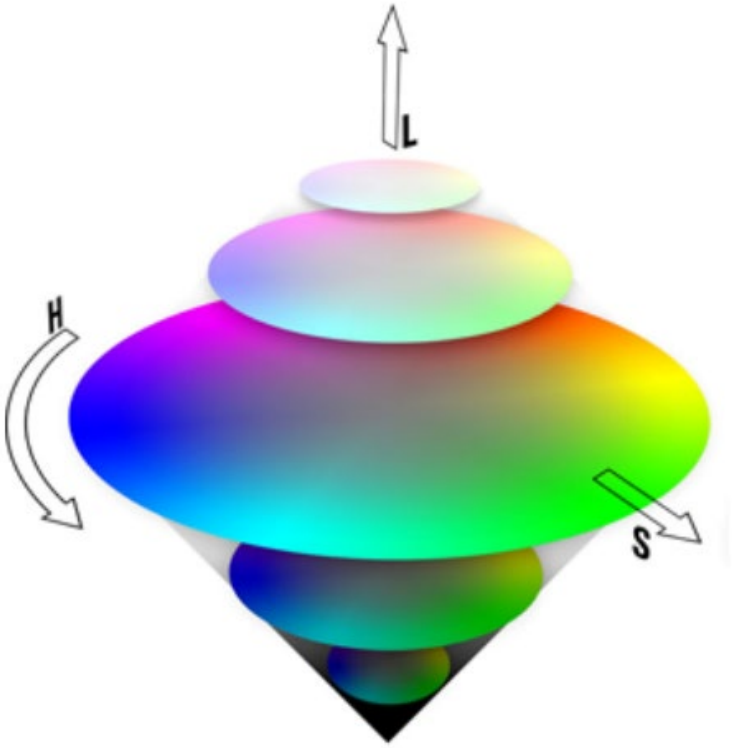
红点的上下移动





红点向圆心方向移动





Hue (color 0-360°)
- Outer edge
- Determined by angle

Saturation (purity 0-1.0)
- Distance from vertical axis
- Closer -> Paler

Intensity (brightness 0-1.0)
- Distance from bottom

Reference: Rafael C. Gonzalez, Richard E. Woods. Digital Image Processing[M]. 3rd Edition

Color Models -- HSI Model

- Converting colors from RGB to HSI

$$H = \begin{cases} \theta & \text{if } B \le G \\ 360 - \theta & \text{if } B > G \end{cases}$$

$$\theta = \cos^{-1}\left\{ \frac{\frac{1}{2}\left[(R-G)+(R-B)\right]}{\left[(R-G)^2+(R-B)(G-B)\right]^{\frac{1}{2}}} \right\}$$

$$S = 1 - \frac{3}{(R+G+B)}\left[\min(R,G,B)\right] \qquad I = \frac{1}{3}(R+G+B)$$

6.3 Pseudocolor Image Processing
Intensity Slicing

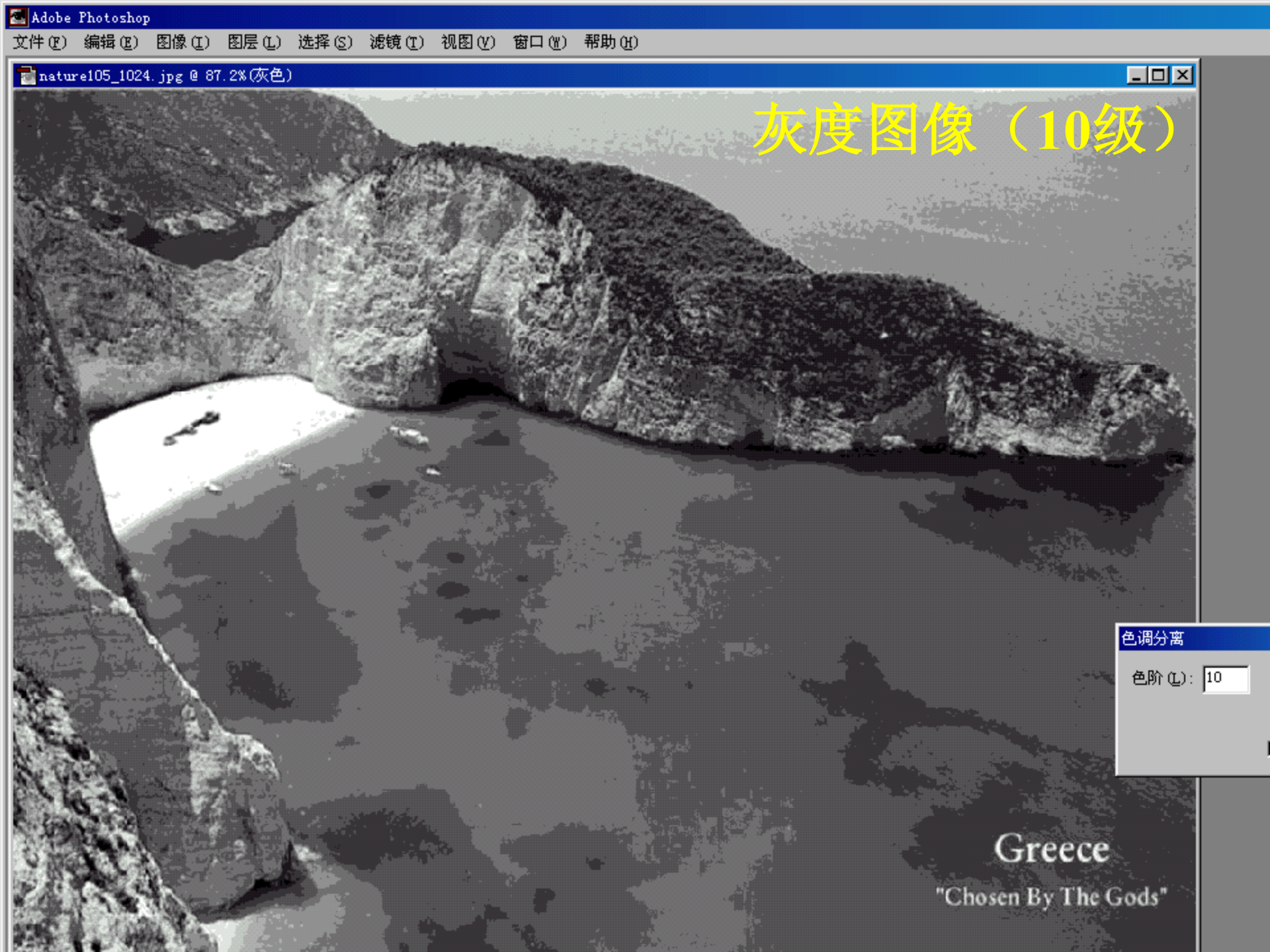
- The **pseudocolors**（伪彩色） is used to distinguish the process of assigning colors to grayscale images.

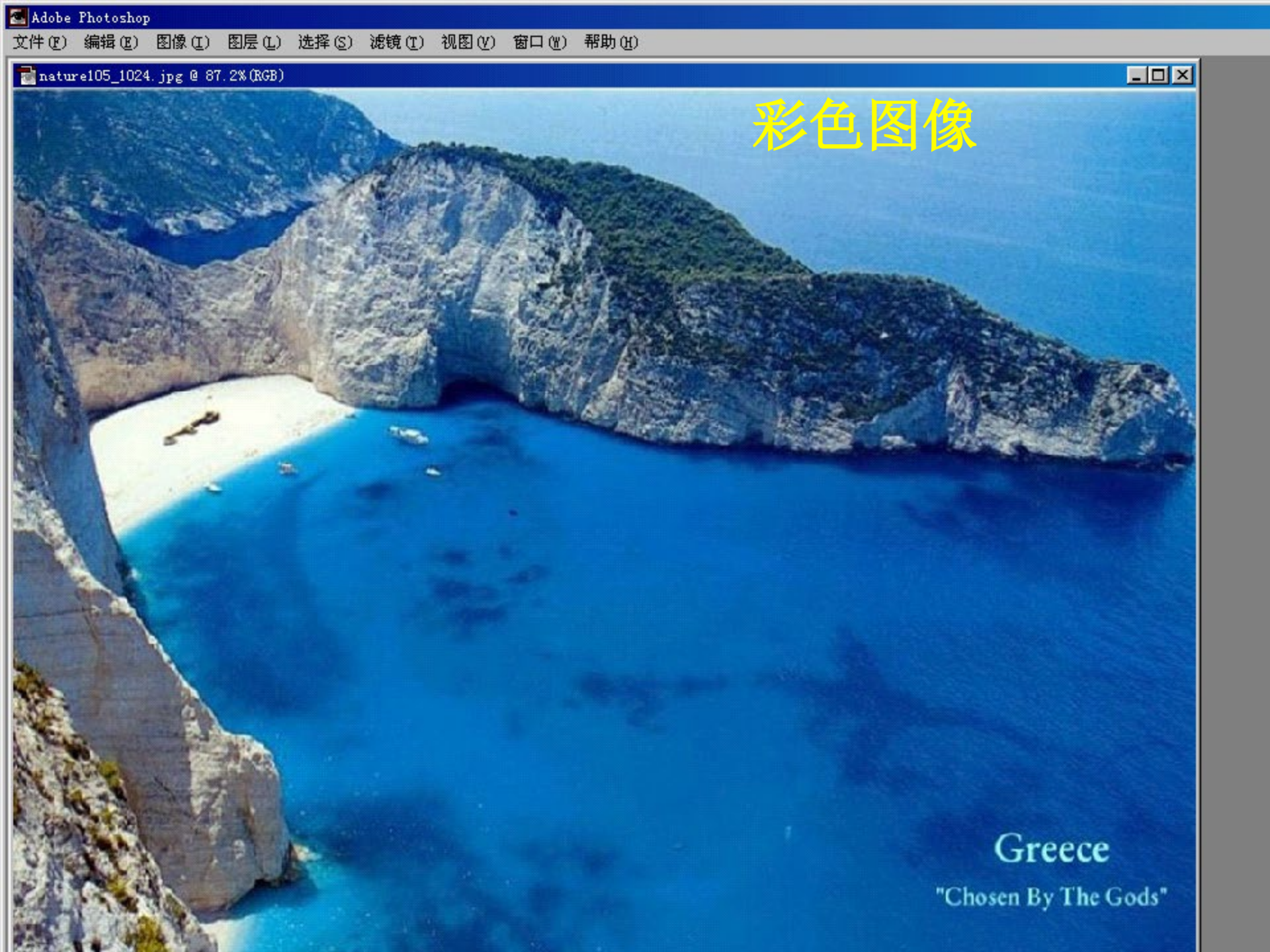
- The technique of **intensity(or density) slicing** and color coding is one of the simplest examples of pseudocolor image processing.

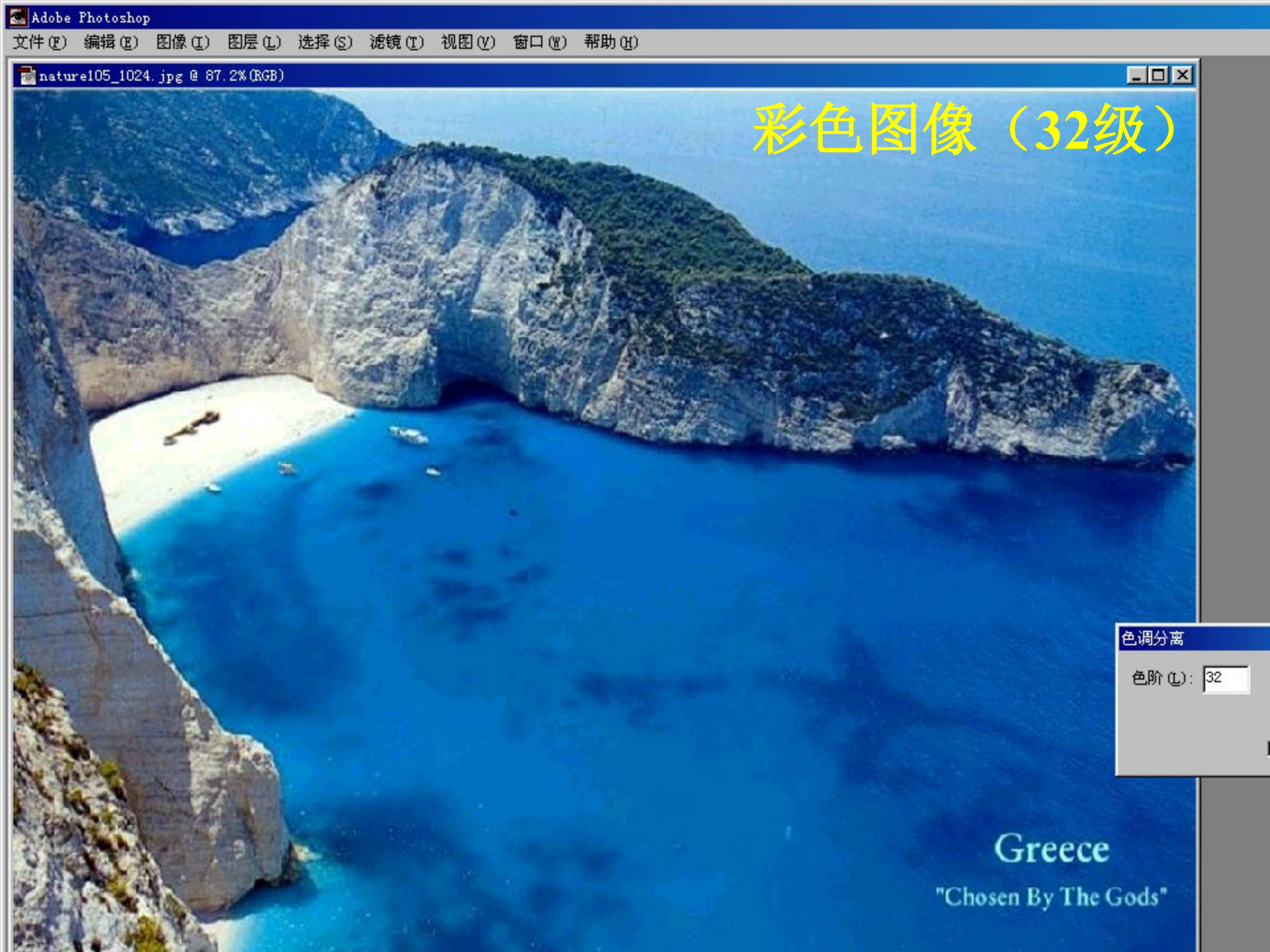

Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(灰色)
灰度图像（255级）
色调分离
色阶(L): 255
Greece
"Chosen By The Gods"

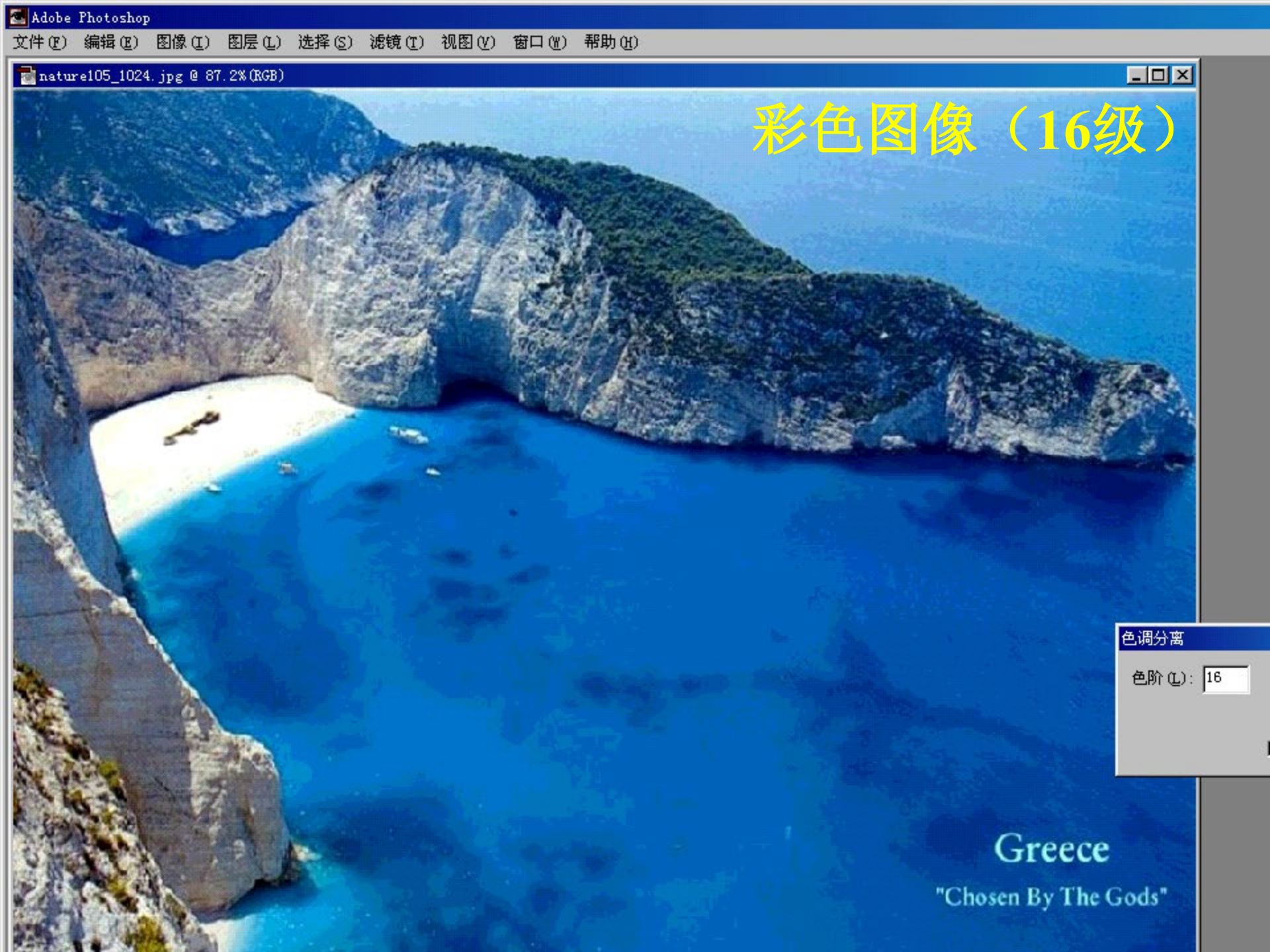

Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(灰色)
灰度图像（50级）
色调分离
色阶(L): 50
Greece
"Chosen By The Gods"

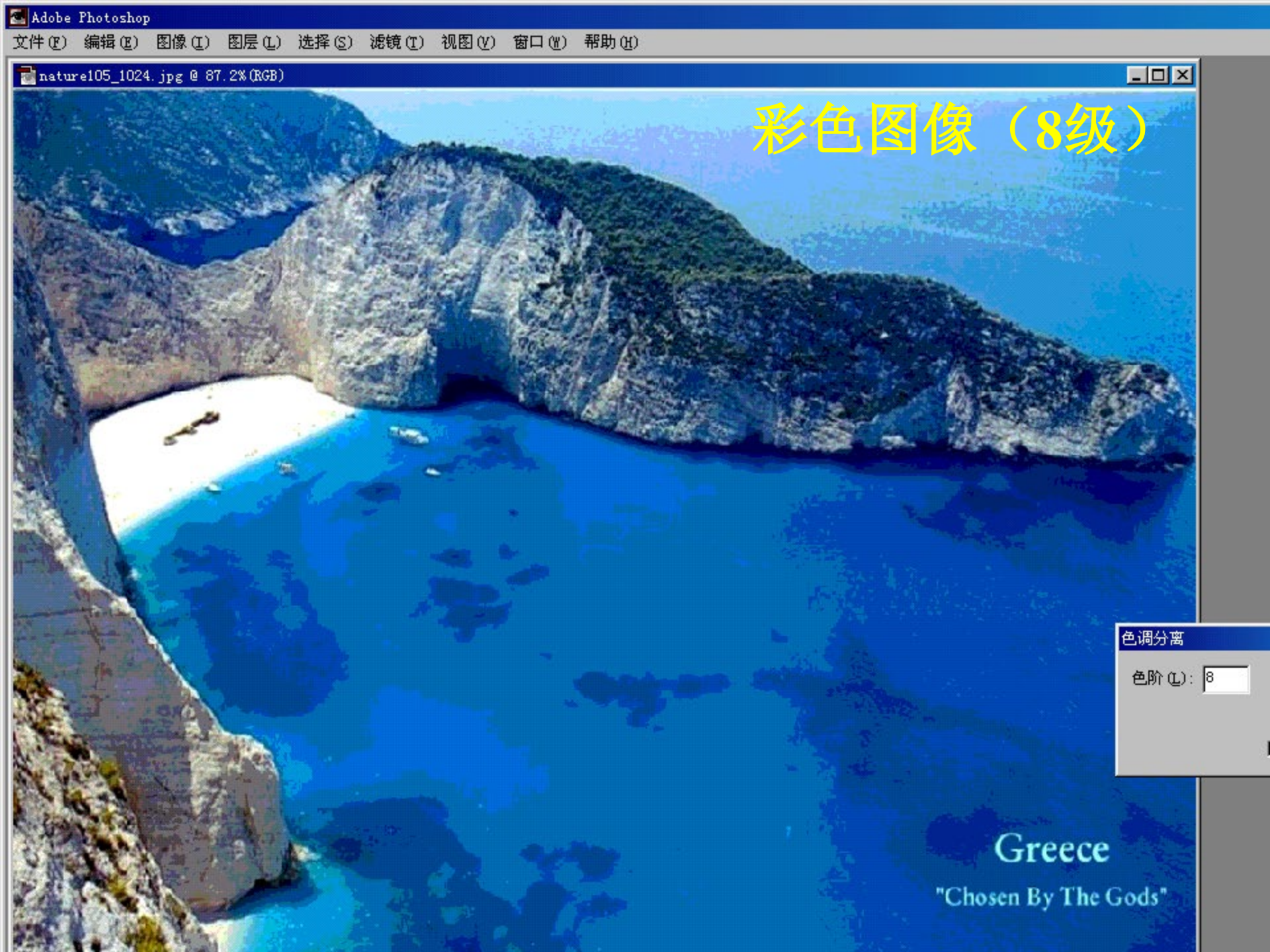

Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(灰色)
灰度图像（30级）
色调分离
色阶(L): 30
Greece
"Chosen By The Gods"

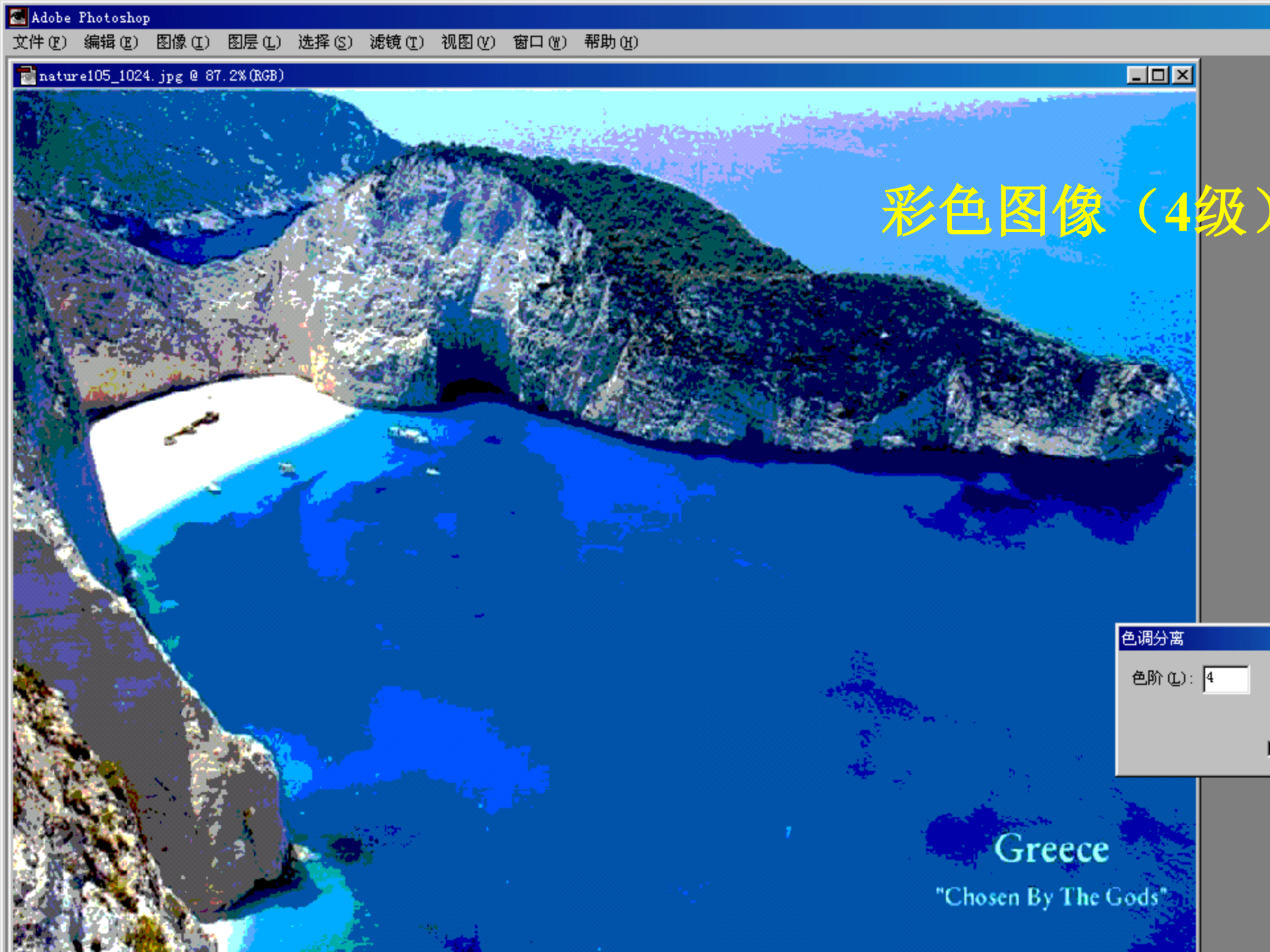

Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(灰色)
灰度图像（10级）
色调分离
色阶(L): 10
Greece
"Chosen By The Gods"


Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(RGB)
彩色图像
Greece
"Chosen By The Gods"


Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(RGB)
彩色图像（32级）
色调分离
色阶(L): 32
Greece
"Chosen By The Gods"


Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(RGB)
彩色图像（16级）
色调分离
色阶(L): 16
Greece
"Chosen By The Gods"


Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(RGB)
彩色图像（8级）
色调分离
色阶(L): 8
Greece
"Chosen By The Gods"


Adobe Photoshop
文件(F) 编辑(E) 图像(I) 图层(L) 选择(S) 滤镜(T) 视图(V) 窗口(W) 帮助(H)
nature105_1024.jpg @ 87.2%(RGB)
彩色图像（4级）
色调分离
色阶(L): 4
Greece
"Chosen By The Gods"

Intensity Slicing

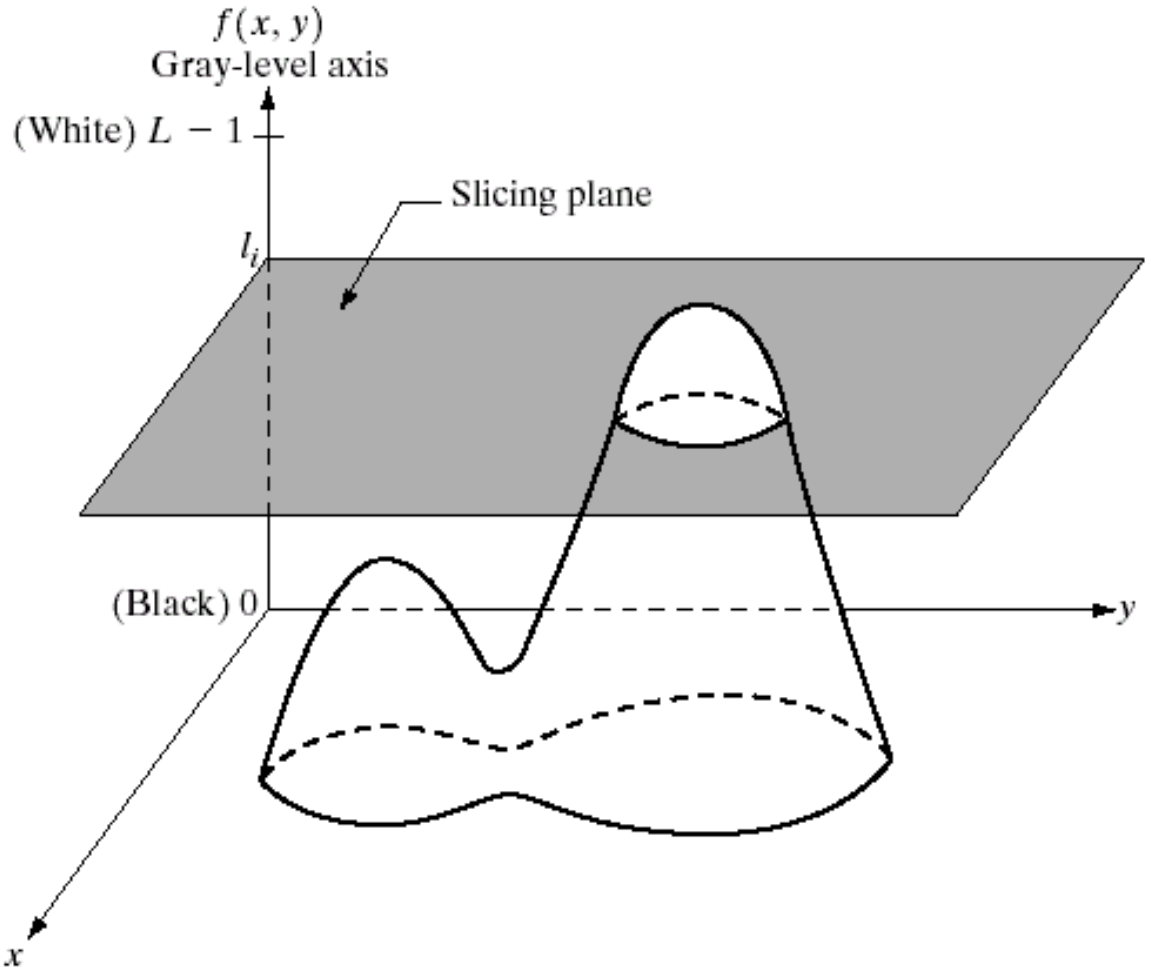


FIGURE 6.18 Geometric interpretation of the intensity-slicing technique.

灰度分层法

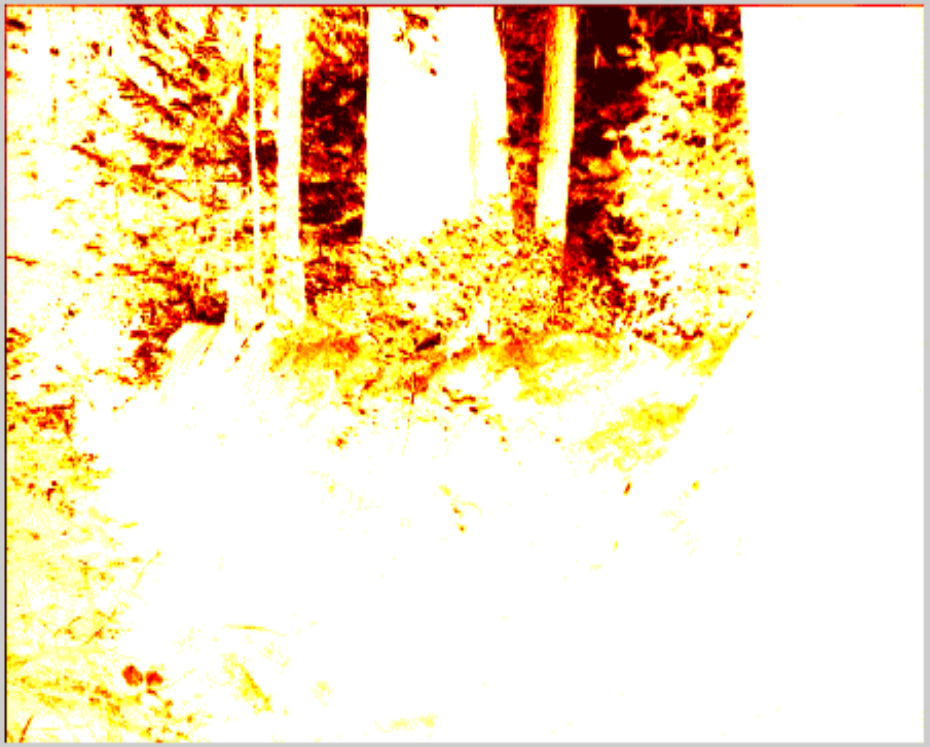
$$f(x,y)=C_i,\quad l_{i-1}\le f(x,y)\le l_i,\quad i=1,2,\cdots,k$$

grayslice

```
I=imread('forest.tif');
x=grayslice(I,16);
figure,imshow(x,hot(16));
```







intensity to color transformations

2. To perform three independent transformations on the intensity of any input pixel. The three results are then converted to the red, green, and blue channels to produce a composite image.

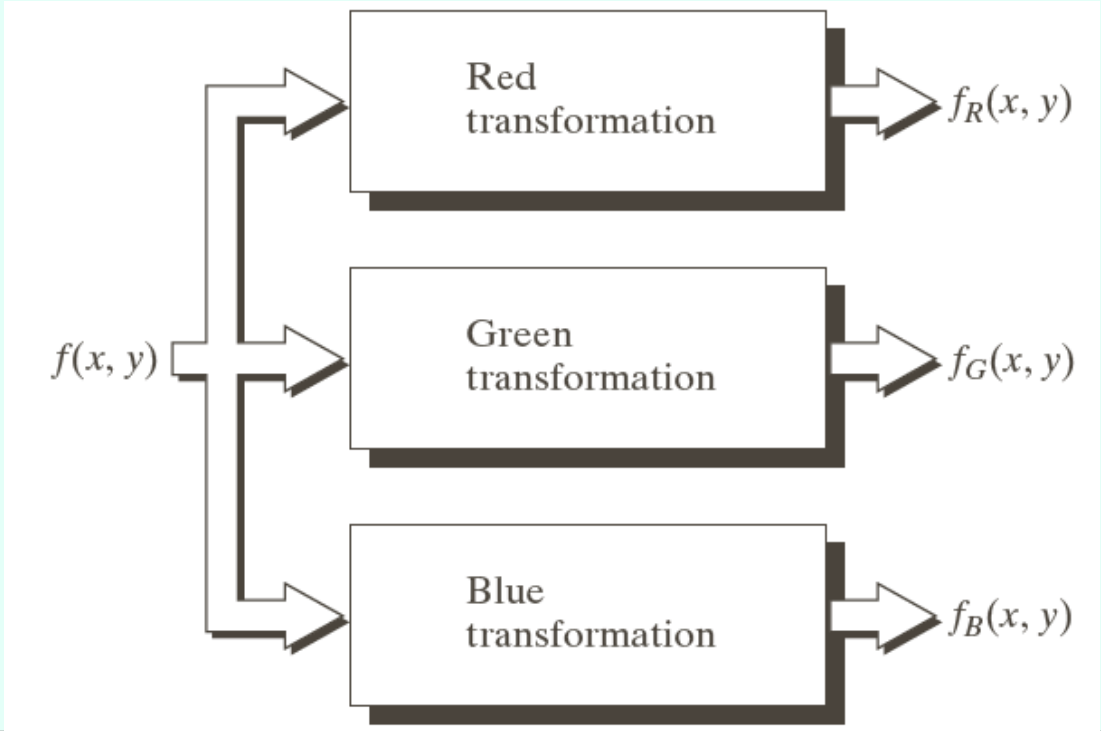
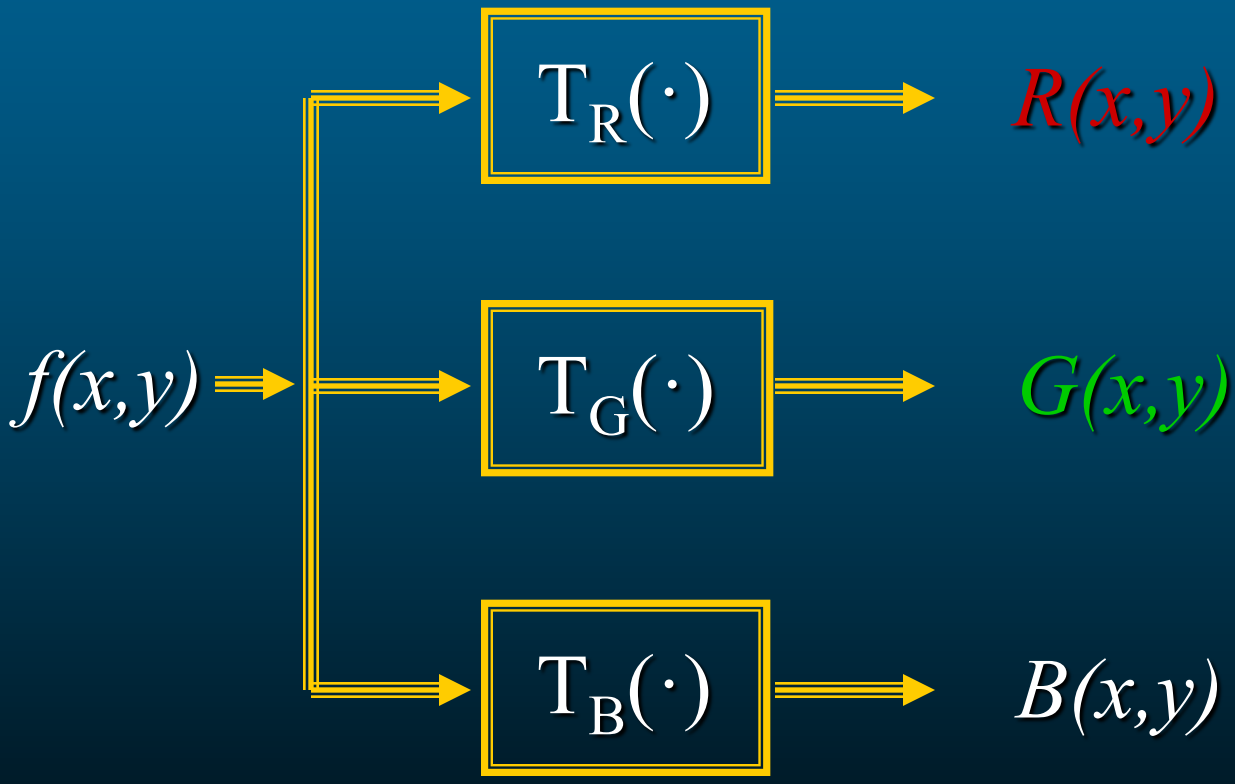
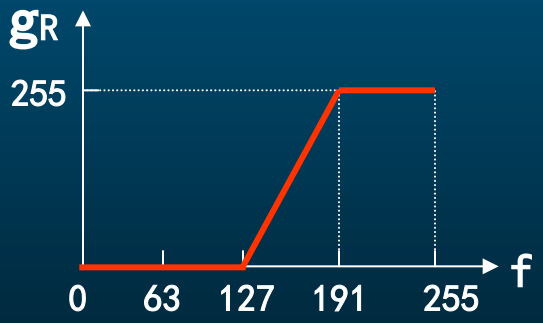


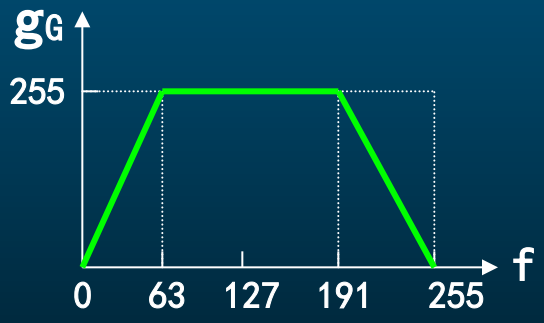
FIGURE 6.23 Functional block diagram for pseudocolor image processing. f_R, f_G, and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

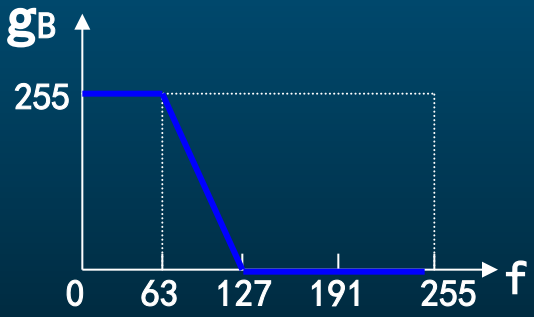
灰度变换法示意图

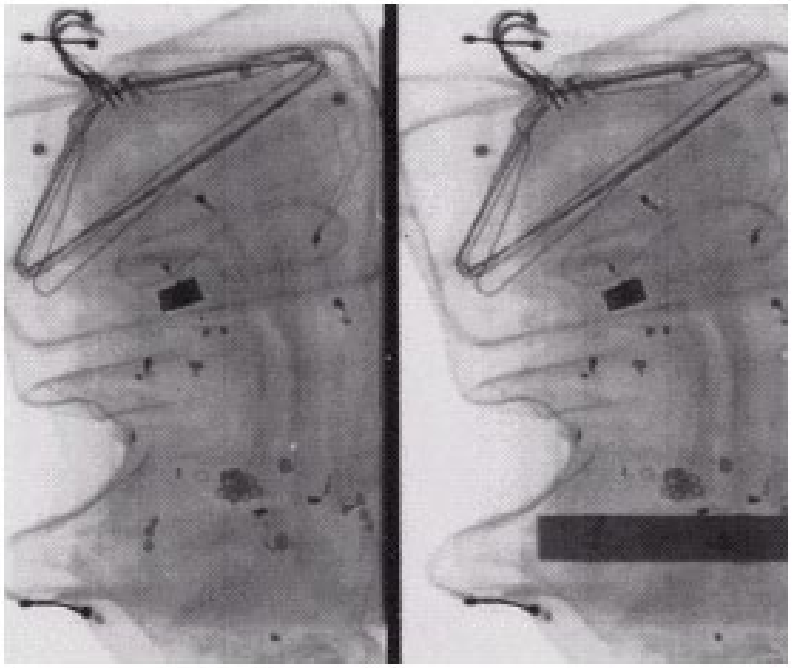


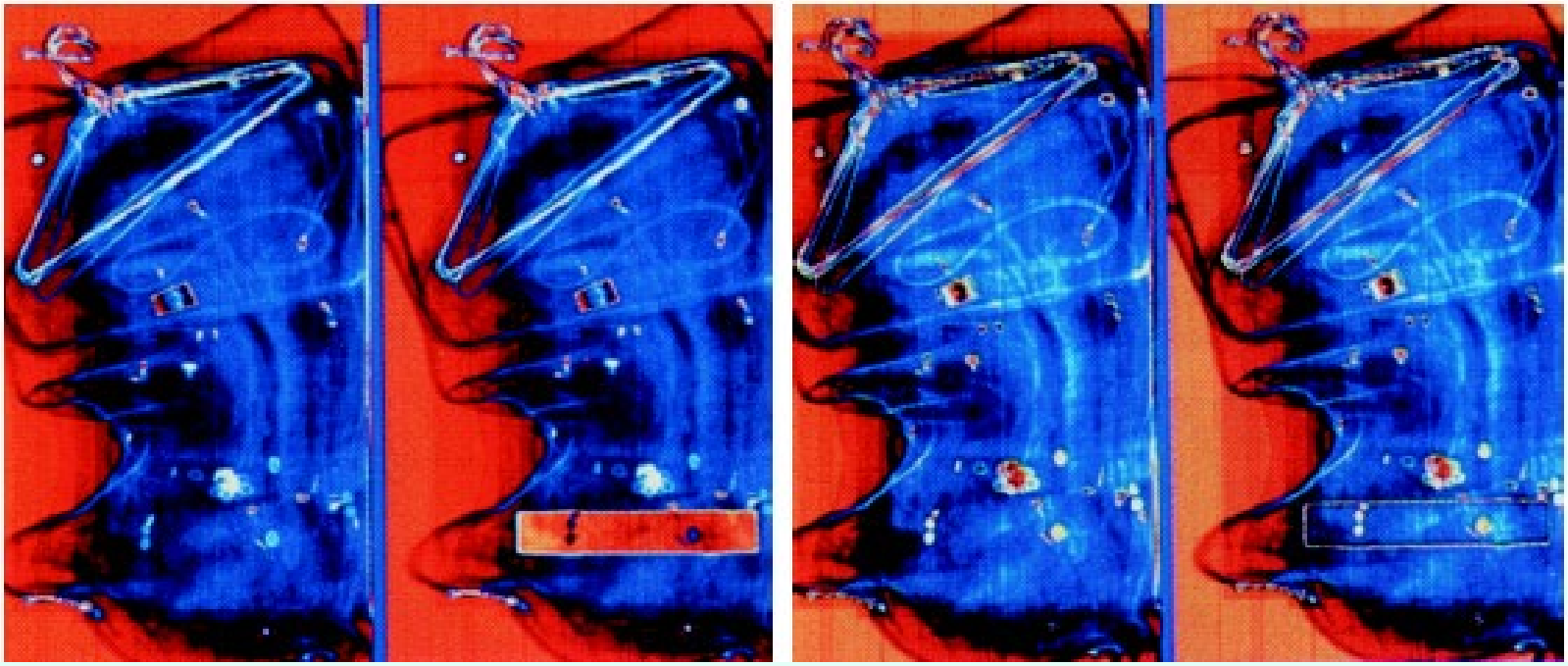
伪彩色处理 灰度变换曲线







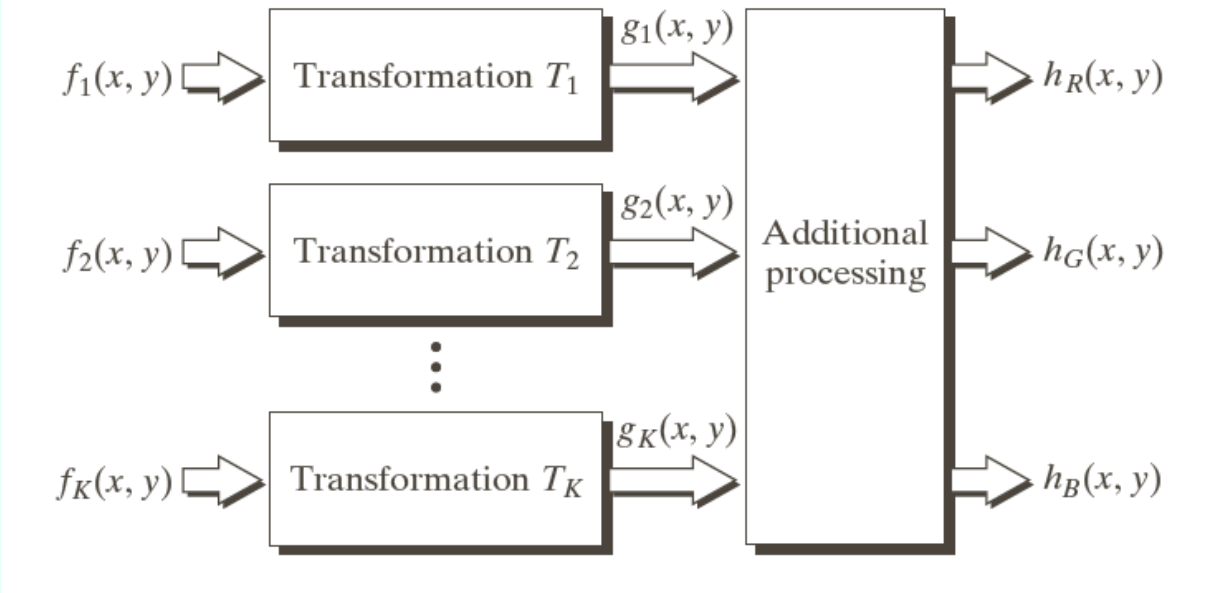


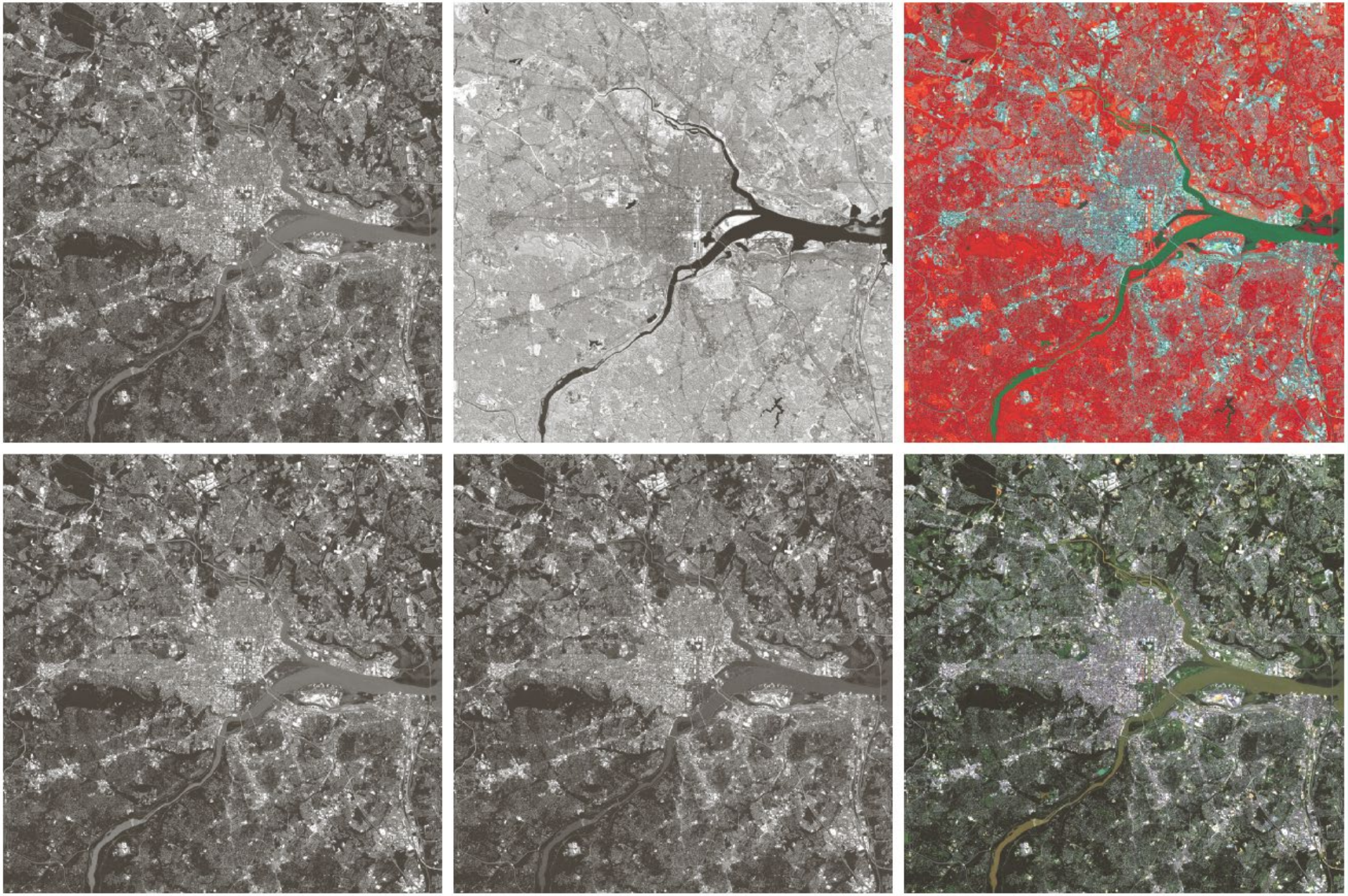


intensity to color transformations

3.To combine several grayscale images into a single color composite.

A frequent use of this approach is in multispectral image process, where different sensors produce individual grayscale images, each in a different spectral band.



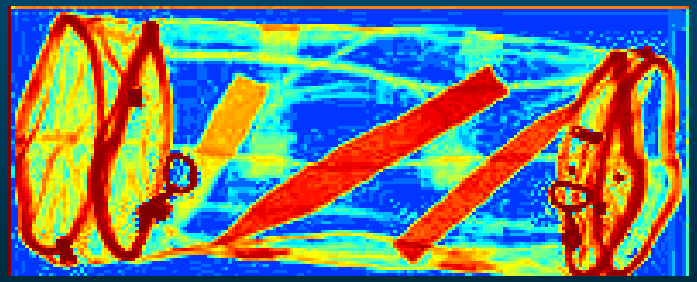




原图

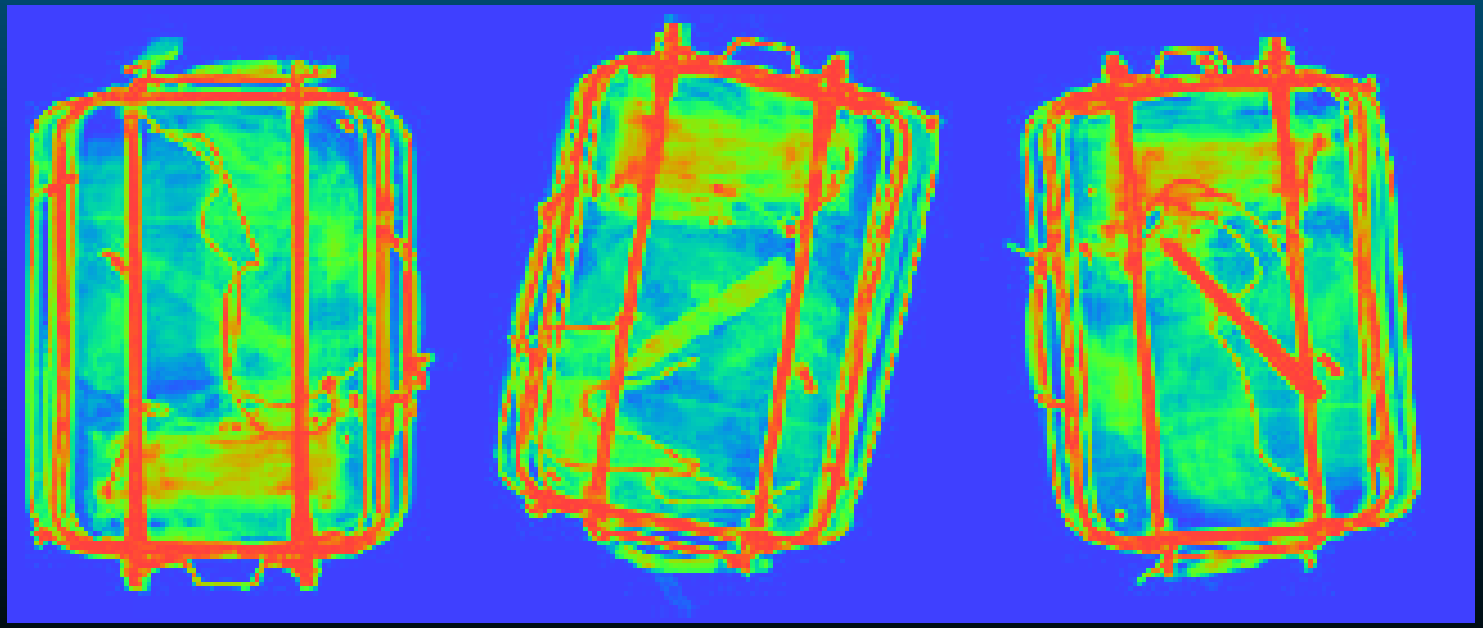


直方图均衡处理图



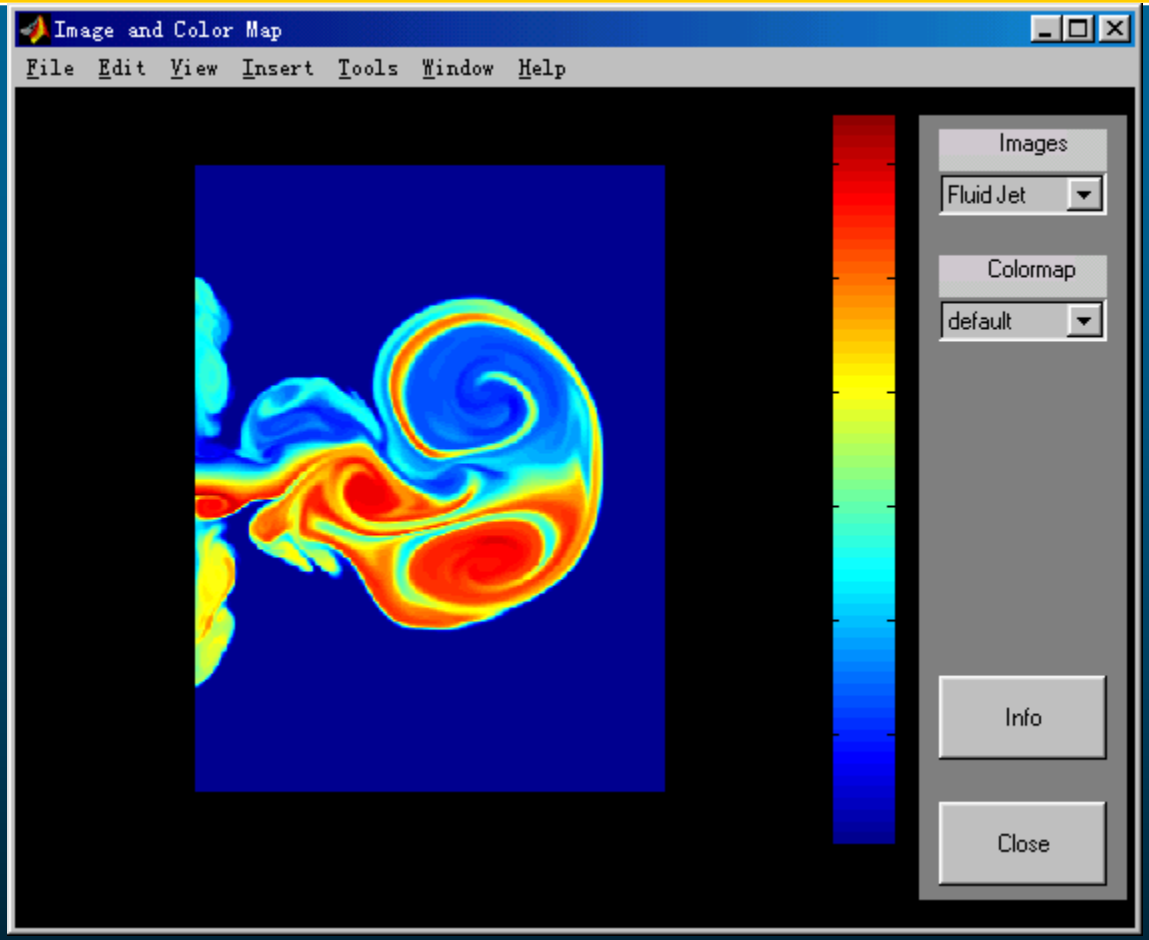
JET伪彩色图

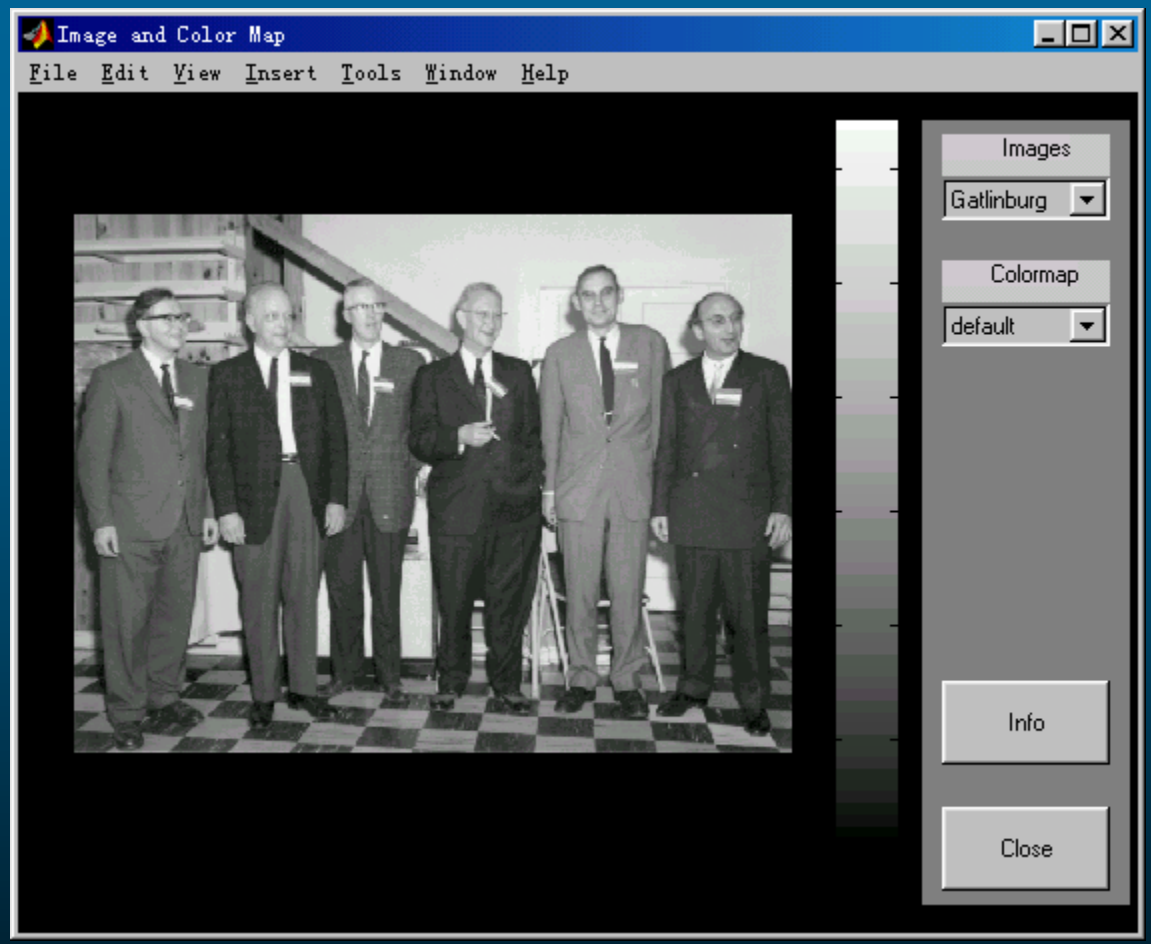


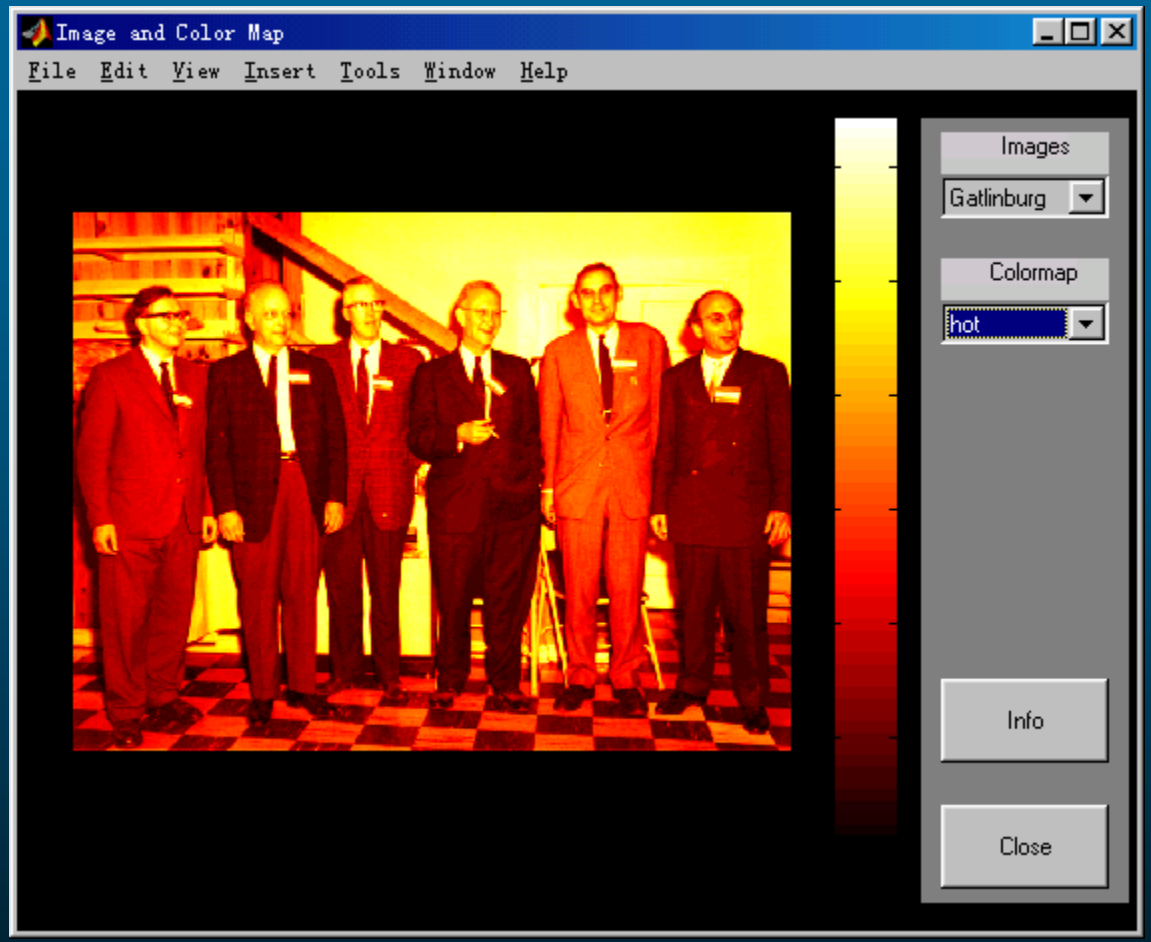


基于亮度表示的伪彩色效果图




Image and Color Map
File Edit View Insert Tools Window Help
Images
Fluid Jet
Colormap
default
Info
Close


Image and Color Map
File Edit View Insert Tools Window Help
Images
Gatlinburg
Colormap
default
Info
Close


Image and Color Map
File Edit View Insert Tools Window Help
Images
Gatlinburg
Colormap
hot
Info
Close



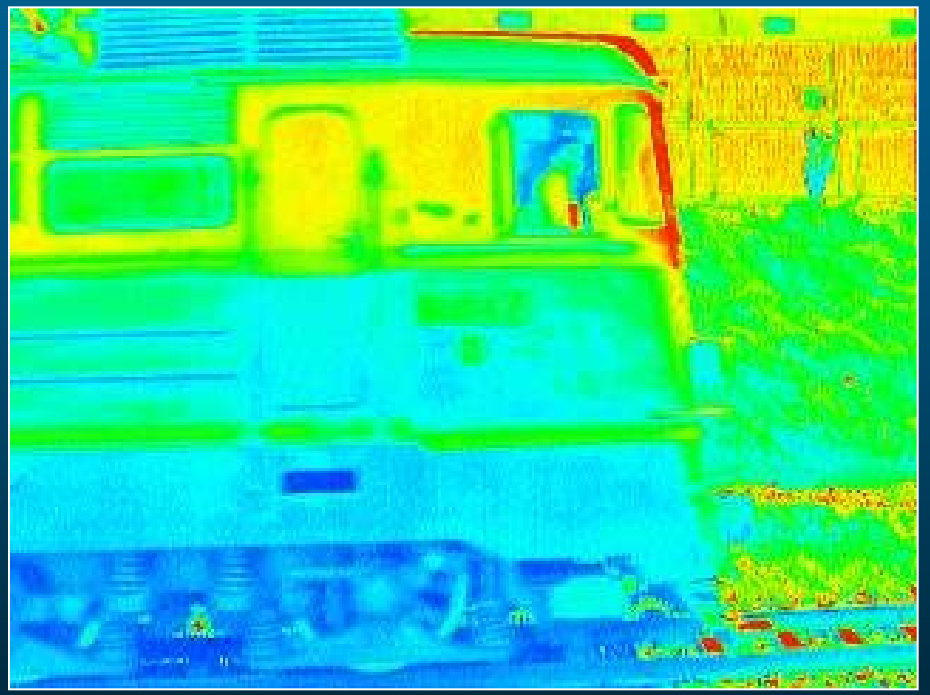


Image Colorization

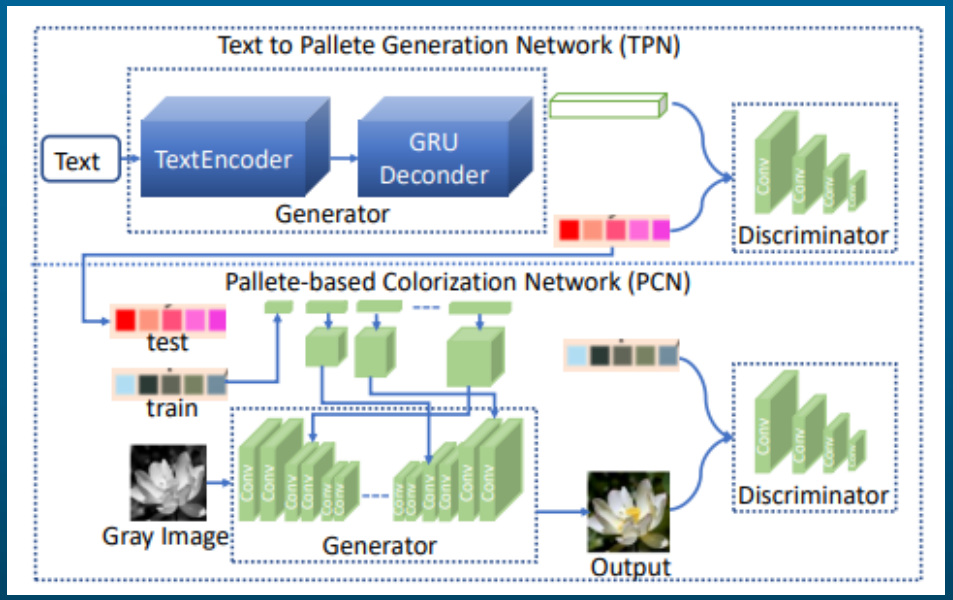




Image Colorization: A Survey and Dataset

BigColor:Colorization using a Generative Color Prior for Natural Images

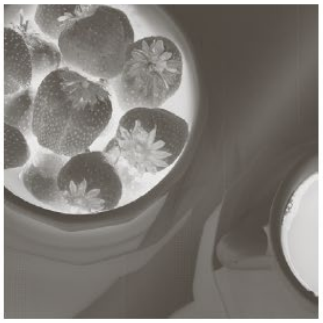
6.4 Full Color Image Processing

- Two processing methods:
  - (1) process each channel (or color component) separately, as if the color image were three gray scale images;
  - (2) process all channels with each pixel represented as a vector.



Full color

FIGURE 6.30 A full-color image and its various color-space components. Interactive.)



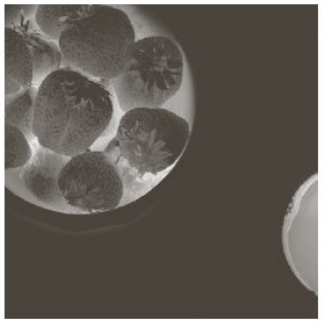
Cyan



Magenta



Yellow



Black

CMYK



Red



Green



Blue

RGB



Hue



Saturation



Intensity

HSI

6.5 Tone and Color Corrections
Tone Corrections

FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not alter the image hues.



Tone and Color Corrections
Tone Corrections

FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not alter the image hues.



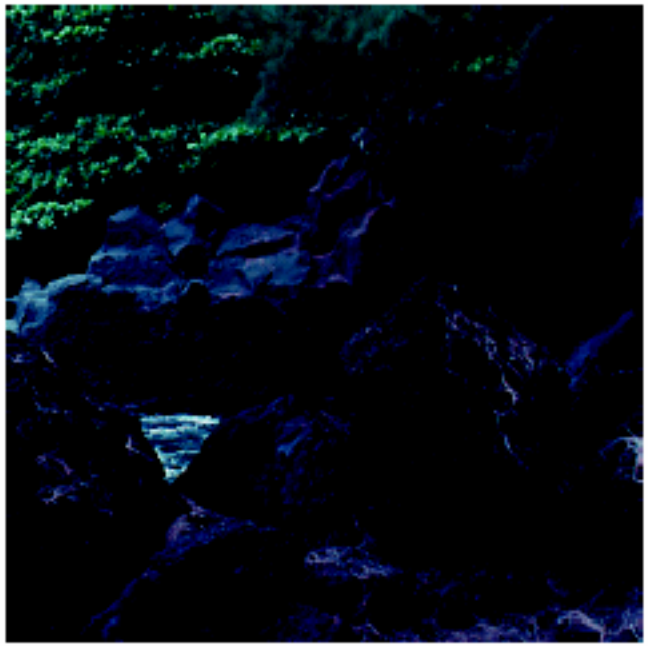
Light



Corrected

Tone and Color Corrections
Tone Corrections

FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not alter the image hues.



Dark



Corrected

6.6 Smoothing and Sharpening
Color Image Smoothing



a b
c d

FIGURE 6.38
(a) RGB image.
(b) Red component image.
(c) Green component.
(d) Blue component.



Smoothing and Sharpening
Color Image Smoothing

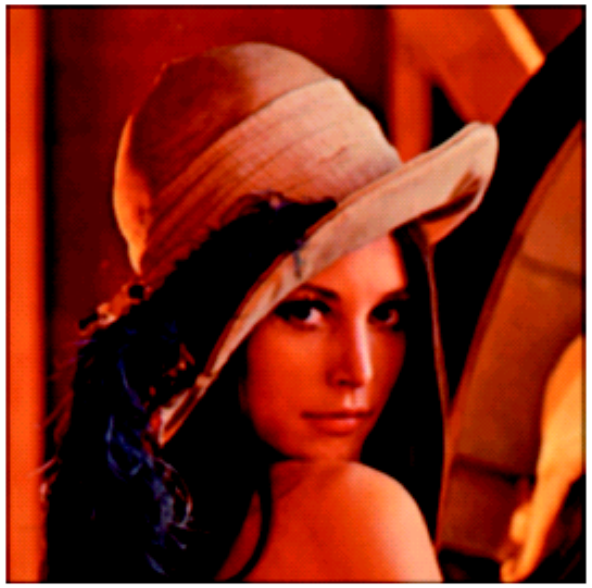


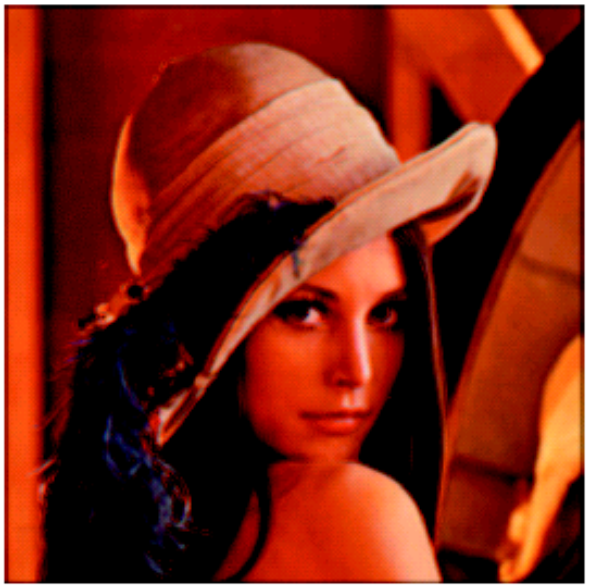
a b c

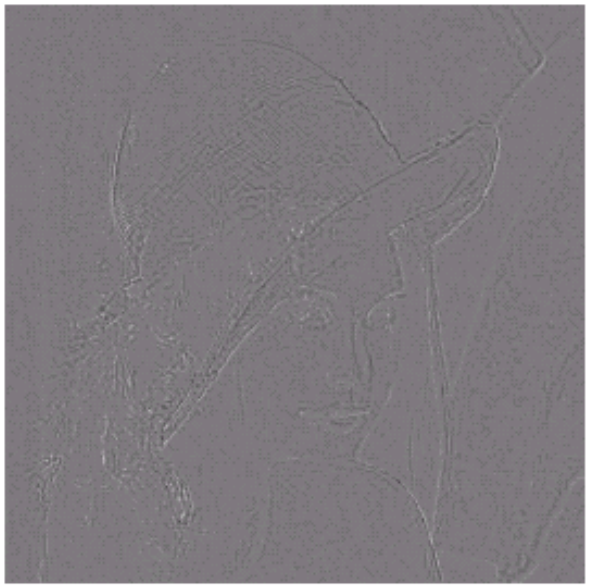
FIGURE 6.39 HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.

Smoothing and Sharpening
Color Image Smoothing



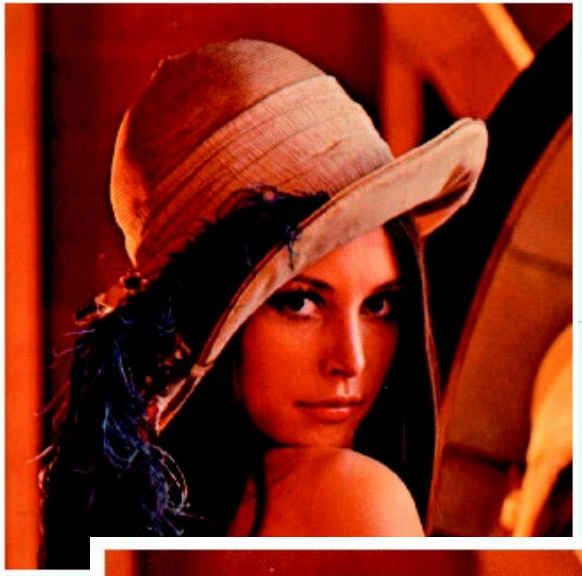






a b c

FIGURE 6.40 Image smoothing with a 5 × 5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.



Smoothing and Sharpening
Color Image Sharpening

- Laplacian method: see Section 3.6.2 (Pages 182-184)



a b c

FIGURE 6.41 Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the intensity component and converting to RGB. (c) Difference between the two results.

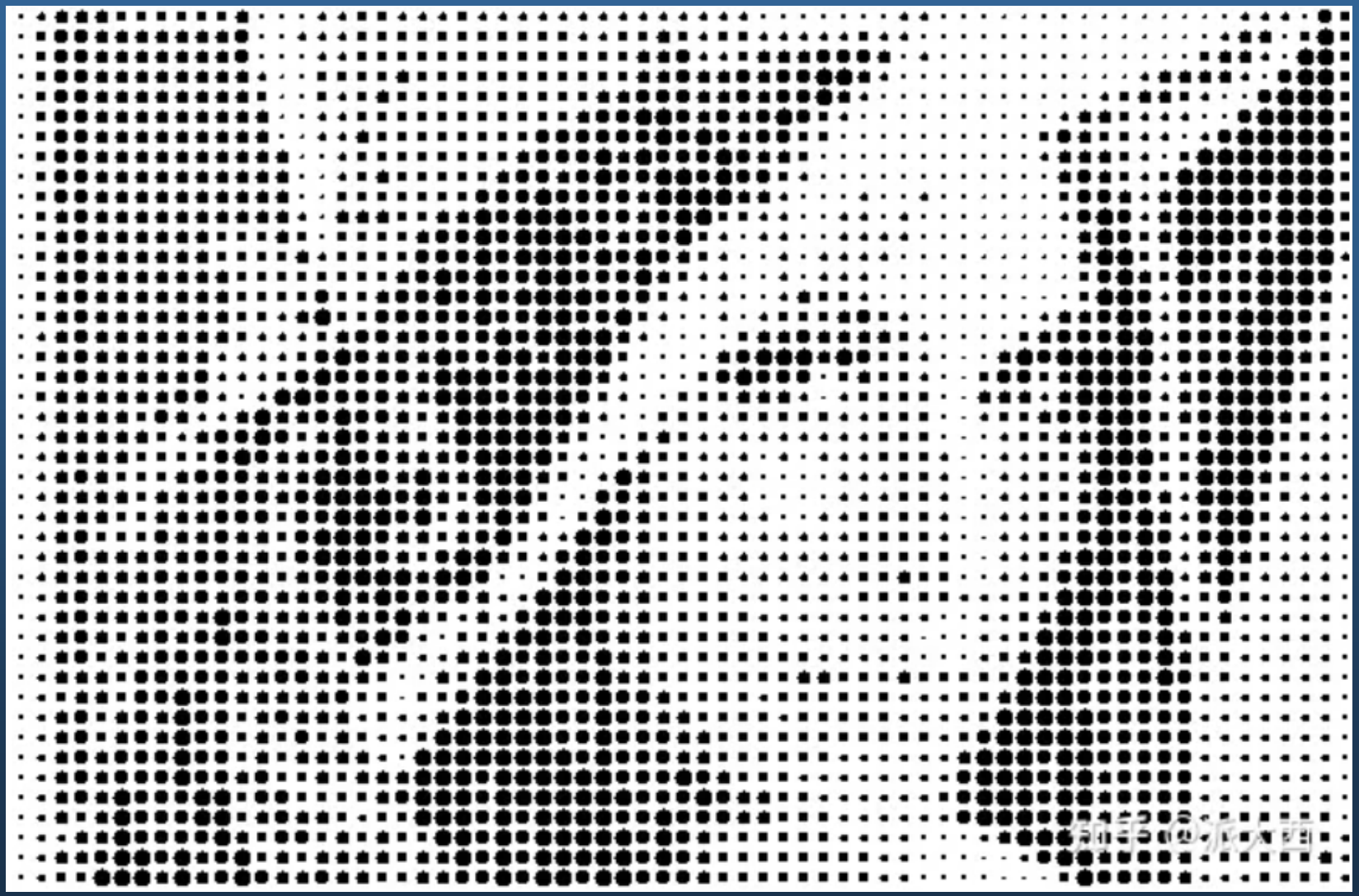
半调（halftoning）技术 - 扩展学习

利用人眼的视觉“错觉”，采用非连续的点模拟连续效果。是一种印刷技术，可节省油墨。

用墨点的大小或频率的改变来模拟阴暗的变化。在一定的距离观察下，便可以使二值化影像重现连续调的感觉









（a）连续色调图像



（b）半调图像

报纸上的图片为100DPI
书或杂志上的图片为300DPI

抖动技术（dithering）：将原始图像 f(x,y) 加上一随机小噪声 d(x,y)，可减少由于灰度级数过少产生虚假轮廓，同时引入噪声。



im2bw(I)

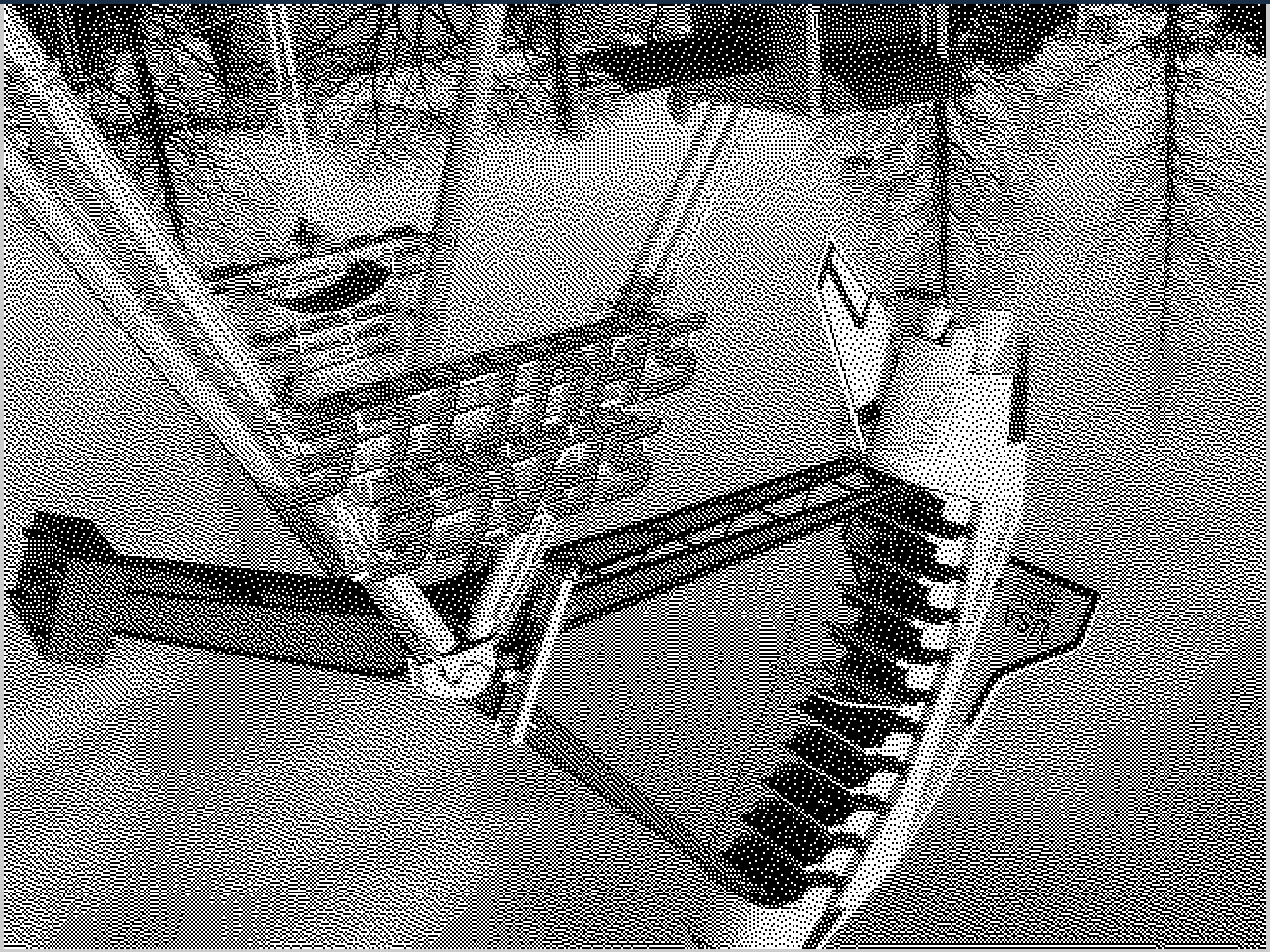


J=imnoise(I,'gaussian',0,0.02);

im2bw(J)




USA
United States
NASA
SPACE
SHUTTLE
2005 2 17



单选题 25分

设置

The hardware-oriented models most commonly used in practice are the （） model for color monitors

A RGB

B CMY

C HSI

提交

单选题 25分

设置

In the RGB model, the color red denotes as (1,0,0), then the color cyan denotes as ( )

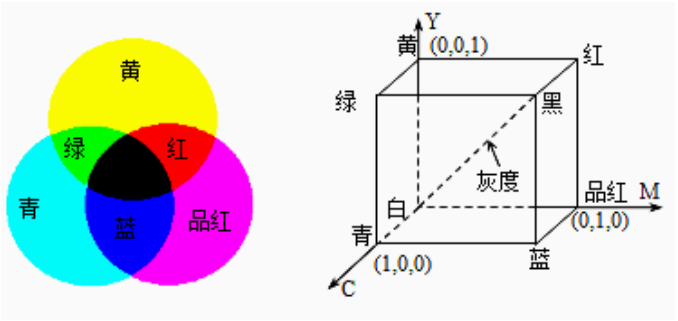
A (1,1,1)

B (0,0,0)

C (0,1,1)

D (0,0,1)

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



提交

填空题 25分

设置

In the HSI color model, H means [填空1]

正常使用填空题需3.0以上版本雨课堂

作答

单选题 25分

设置

Pseudocolor image processing consists of assigning colors to gray values based on a specified criterion, we can assign same color to different gray values. Whether this statement is true?

A true

B wrong

提交

Chapter 5
Image Restoration

Chapter 5: Image Restoration

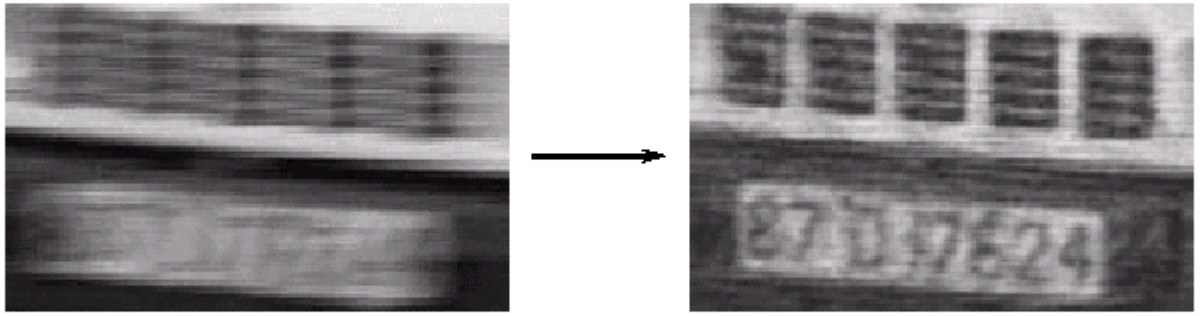
- Image enhancement is largely a subjective process, while image restoration is for the most part an objective process.

- Restoration attempts to reconstruct or recover an image that has been degraded by using a priori knowledge of the degradation phenomenon.

Chapter 5: Image Restoration

Image restoration

- Restoration of images degraded by bad focusing, motion etc.



Blur caused by motion

After restoration

Image Degradation/Restoration Process Model

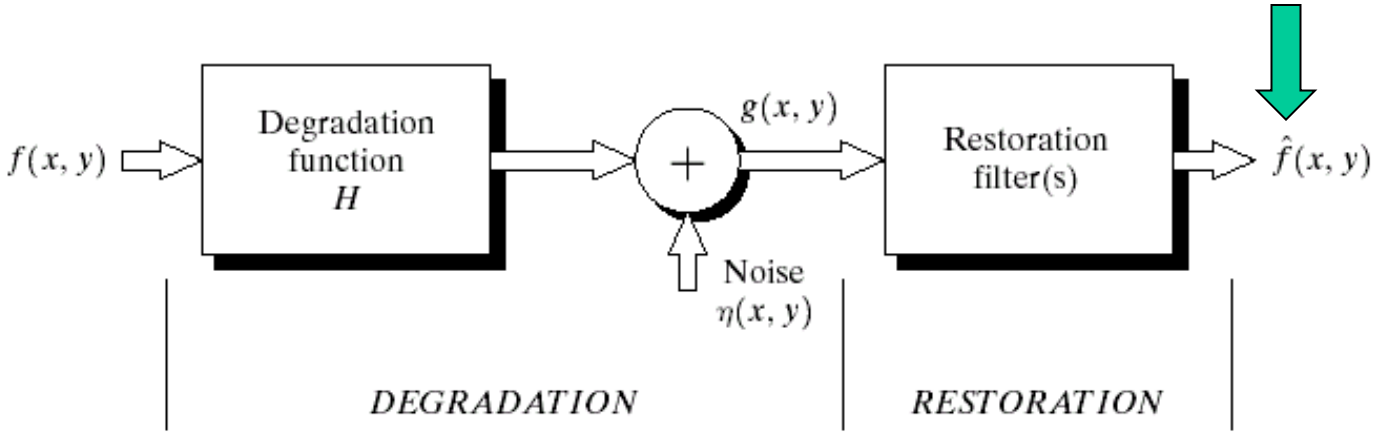
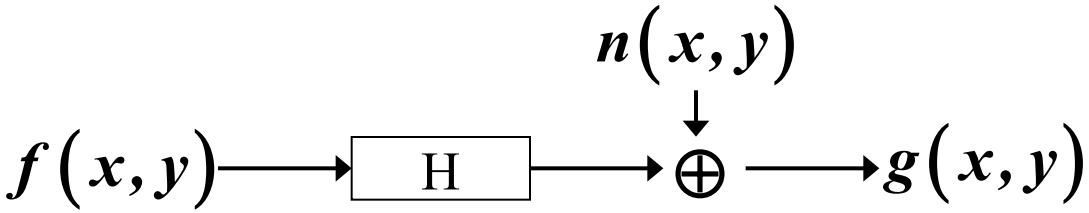


FIGURE 5.1 A model of the image degradation/restoration process.

（1）退化过程



抽象为一个退化系统H以及加性噪声的影响

$$g(x,y)=H\left[f(x,y)\right]+n(x,y)$$

$$H\left[f(x,y)\right]=f(x,y)*h(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}f(\alpha,\beta)h(x-\alpha,y-\beta)d\alpha d\beta$$

（2）退化模型

$$g(x,y)=f(x,y)*h(x,y)+n(x,y)$$

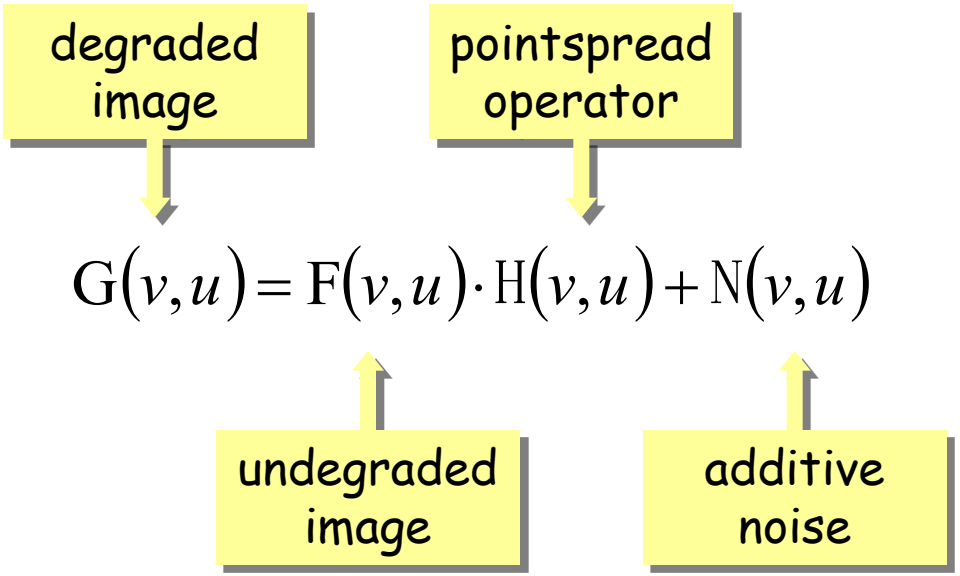
$h(x,y)$ 称为点扩散函数（PSF），其傅里叶变换$H(u,v)$也称为光学传递函数（OTF）

图像复原是指在给定退化图像 $g(x,y)$，了解退化的点扩散函数$h(x,y)$和噪声项 $n(x,y)$ 的情况下，估计出原始图像 $f(x,y)$

Image Degradation Model

(Frequency Domain)

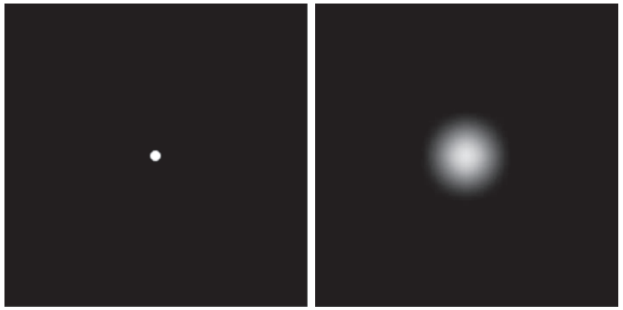
$$g(x,y) = f(x,y) * h(x,y) + n(x,y)$$



Estimating the Degradation Function

- Observation $H_s(u,v) = \dfrac{G_s(u,v)}{\hat{F}_s(u,v)}$

- Experimentation $H(u,v) = \dfrac{G(u,v)}{A}$



- Mathematical modeling $H(u,v) = e^{-k(u^2+v^2)5/6}$

Inverse filter （逆滤波）

The linear degradation model

$$g(x,y) = f(x,y) * h(x,y) + n(x,y)$$

$$G(u,v) = F(u,v)H(u,v)$$ 不考虑噪声

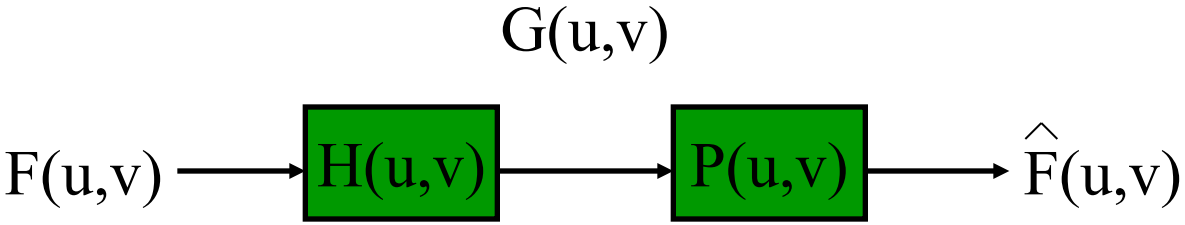
$$\hat{F}(u,v) = G(u,v) / H(u,v)$$

$$\hat{f}(x,y) = \mathrm{F}^{-1}[G(u,v) / H(u,v)]$$

$$P(u,v) = \frac{1}{H(u,v)}$$ 恢复与退化过程点扩散函数互为倒数

Inverse filter （逆滤波）

当考虑噪声



$$G(u,v) = F(u,v)H(u,v) + N(u,v)$$

$$P(u,v) = \frac{1}{H(u,v)}$$

恢复函数

$$\hat{F}(u,v) = G(u,v)P(u,v)$$
$$= F(u,v) + N(u,v)/H(u,v)$$

Inverse filter 逆滤波复原



模糊图像

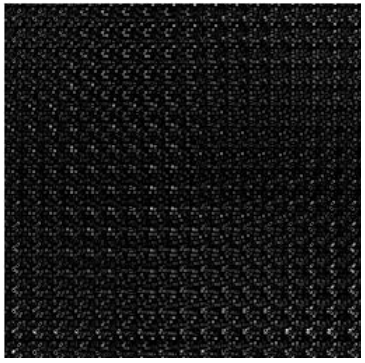


逆滤波



模糊加噪声





直接逆滤波

Minimum Mean Square Error Filtering -wiener filtering

The objective is to find an estimate $\hat{f}$ of the uncorrupted image f such that the mean square error between them is minimized.

准则：

$$\min: \quad e^2 = E\left\{\left[f(x,y) - \hat{f}(x,y)\right]^2\right\}$$

求 $\hat{f}(x,y)$。

维纳滤波复原 （Wiener filtering）

原理

- 一种有代表性的约束复原方法，使原始图像和复原图像之间均方误差最小的复原方法。

均方误差： $e^2 = E\left[\left(f - \hat{f}\right)^2\right]$

恢复函数：
$$H_w(u,v) = \frac{1}{H(u,v)} \cdot \frac{|H(u,v)|^2}{|H(u,v)|^2 + S_n(u,v)/S_f(u,v)}$$

功率谱： $S_f(u,v) = |F(u,v)|^2 \quad S_n(u,v) = |N(u,v)|^2$

噪声为零，噪声功率谱小，维纳滤波即为逆滤波；实际问题中，功率谱未知，用常数K来代替二者的比值

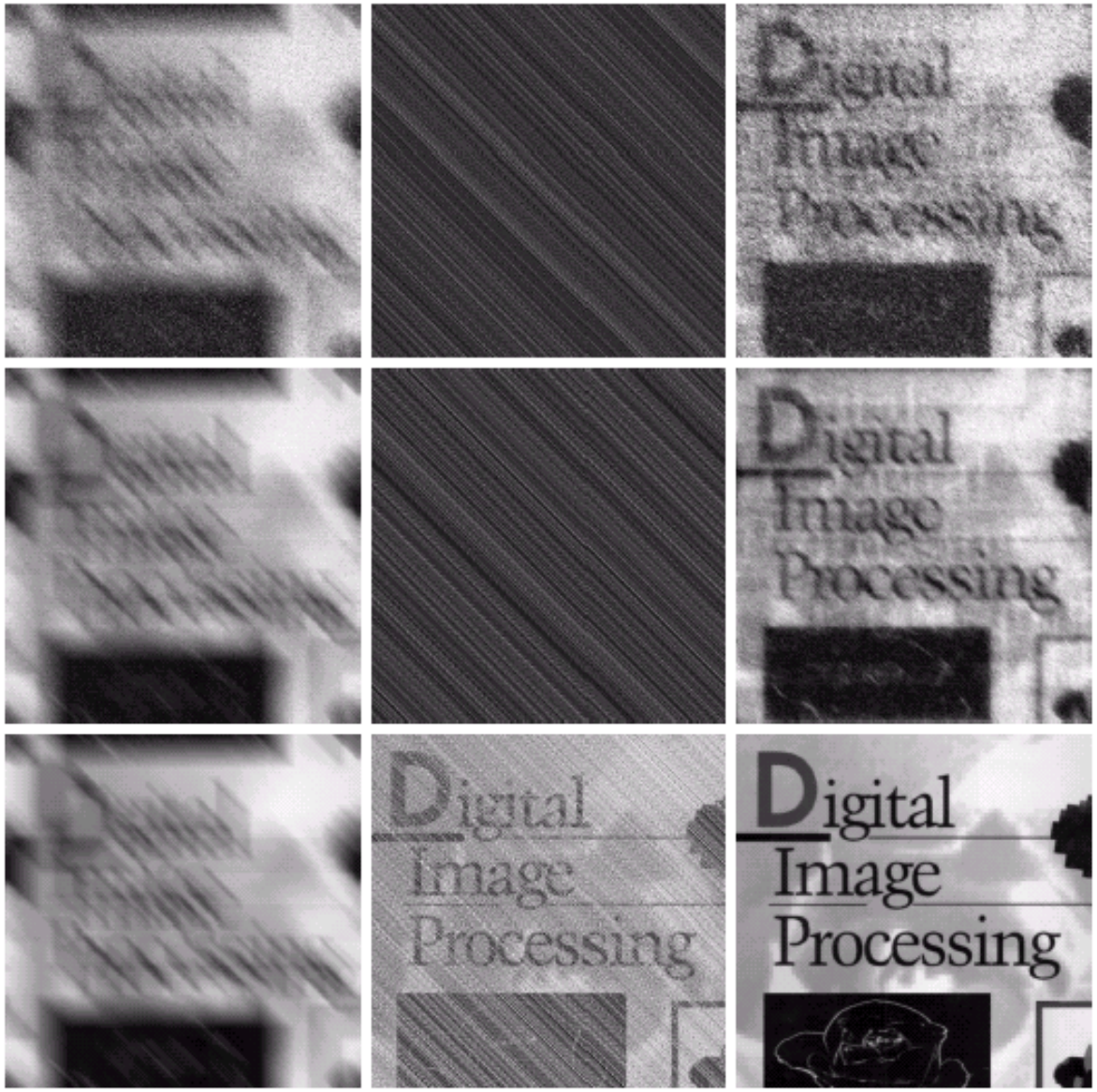
Minimum Mean Square Error Filtering -wiener filtering

As the SNR is unknown, we can substitute it by a constant K

$$\hat{F}(u,v) = \left[\frac{1}{H(u,v)} \bullet \frac{|H(u,v)|^2}{|H(u,v)|^2 + K}\right] G(u,v)$$

复原后图像

MATLAB function： **deconvwnr**



a b c
d e f
g h i

FIGURE 5.29 (a) Image corrupted by motion blur and additive noise. (b) Result of inverse filtering. (c) Result of Wiener filtering. (d)–(f) Same sequence, but with noise variance one order of magnitude less. (g)–(i) Same sequence, but noise variance reduced by five orders of magnitude from (a). Note in (h) how the deblurred image is quite visible through a "curtain" of noise.

Minimum Mean Square Error Filtering -wiener filtering